# Contents

# 1 The ordered sound-change sequence

## 1.1 Scope and orientation

The sequence begins with early West Germanic consonant and vowel changes and ends with Old English r-metathesis.

Rhotacism, brightening, breaking, umlaut, and apocope alternate with narrowly conditioned changes whose relative order rests on particular witness words.

The evidence ranges from broadly attested sound laws to lexical constraints that establish only one chronological boundary.

## 1.2 Numbering note

SC numbers remain the established legacy identifiers. The Version 1 book presents the changes in historical chapter order, which differs from the computational cascade order for several rules.

SC038, SC062, and SC084 mark technical or prosodic stages rather than sound changes; SC077 is unused.

# 2 Chapter 1. From Proto-Northwest Germanic to Proto-West Germanic

## 2.1 Historical interval

This chapter covers the sound changes that took place in the proto-language shared by the West Germanic languages — Old English, Old Frisian, Old Saxon, Old High German, and Old Dutch — before the individual languages diverged. The starting reconstruction is Proto-Northwest Germanic (PNWGmc), the hypothetical common ancestor of North Germanic and West Germanic together; the ending reconstruction is Proto-West Germanic (PWGmc), the immediate common ancestor of the West Germanic languages specifically.

## 2.2 Scope and internal diversity

Changes in this chapter are not all equally pan-West-Germanic in scope. They may be grouped broadly as follows:

Northwest Germanic innovations (shared by both North and West Germanic): innovations in the unstressed vowel system, certain final-syllable vowel changes, and selected consonant cluster simplifications. Changes labelled `NWGmc` in the CAPR rule names fall here, though rule prefixes are not always reliable guides to historical scope.

Proto-West-Germanic innovations (shared within West Germanic but not in North Germanic): the cluster of morphological and phonological changes that distinguish Old English, Old High German, Old Saxon, and Old Frisian from Old Norse. Changes labelled `PWGmc` in the CAPR rule names generally fall here. They include early apocope rules, certain consonant assimilations, and the West Germanic gemination of consonants before `*j`.

## 2.3 Major changes

The chapter opens with the root-noun nominative `*-z` loss (SC096), the generalization of endingless nominatives through the athematic consonant stems, complete before Proto-West Germanic: none of the West Germanic daughters shows any ending in this class (Ringe and Taylor 2014, 118). It is the earliest of the three historically distinct final-`*z` developments; the other two (SC020 and SC097) open Chapter 2.

The unstressed `*ai > *ē` development (SC014) represents one of the most pervasive shared NW–West Germanic vowel shifts, turning unstressed endings such as the dative singular and strong-adjective plural to longer vowels. Ringe and Taylor treat this as one of the clearest post-PNWGmc shared developments (Ringe and Taylor 2014, 40–41); Fulk groups it among the North/West-Germanic shared innovations that distinguish the period from Gothic (Fulk 2018, sect. 5.2). The corresponding stressed monophthongization (SC004) belongs later in the cascade and is treated in Chapter 2.

The West Germanic consonant changes of this chapter — j-gemination (SC010), early i-apocope (SC006), coronal-w assimilation (SC008), and related rules — represent the most productive phonological territory for the CAPR derivations. They feed a large proportion of the distinctive consonant clusters of Old English. Handbooks vary in exactly how they group and name these changes (Campbell 1959, sects 404, 406; Hogg 1992, sect. 7.1).

The nasal spirant corridor (SC026–SC027), treated in Chapter 2 at its cascade position, illustrates a type of change common in historical grammars of the “Ingvaeonic” or “North Sea Germanic” area: nasals disappear before voiceless fricatives, with compensatory vowel lengthening (Campbell 1959, sects 462–463; Hogg 1992, sect. 7.77). The CAPR model splits this into two ordered steps to make the vowel effect computationally tractable; the book prose explains that split against the handbook tradition, which typically presents the change as a single process.

Chapters in this part of the book follow the executable cascade order, which models the reconstructed chronology itself. Several rules that carry `PWGmc` labels — final bare-`*a` loss (SC041), surviving bimoric `*ō` unrounding (SC042), and Sievers-law syncope (SC050) — execute later in the cascade and are therefore presented in Chapter 3, where their individual sections discuss their historical stage labels. Conversely, one rule with a West Saxon label, the palatal-glide rule (SC016), executes early and is presented in this chapter; its section notes the mismatch, which will be resolved in a later renaming pass.

One historically Proto-Germanic change, Gm-simplification (`SC002 PGmcGmSimplification`), precedes everything in this chapter as a support stage of the cascade. It is documented in the book-entry plan and its literature dossier confirms the source base is narrow (two lexical families: **draugma-* 'dream' and **taugma-* 'team'; (Kroonen 2013, 101, 511)). A reader-facing section for SC002 awaits a stronger explanatory source base and is not yet assembled in the reader-facing sequence.

## 2.4 A note on source terminology and subgrouping

The literature uses several partly overlapping stage labels for this period:

- Northwest Germanic: the node uniting North and West Germanic.
- Proto-West Germanic: the node uniting only the West Germanic languages.
- North Sea Germanic or Ingvaeonic: a proposed subgroup within West Germanic covering Old English, Old Frisian, and Old Saxon (and sometimes Old Low Franconian), sharing certain innovations over a broader area.
- Anglo-Frisian: a narrower proposed subgroup linking only Old English and Old Frisian.

These labels are not always used consistently across sources. Ringe and Taylor are cautious about reconstructing a discrete Proto-West-Germanic node (Ringe and Taylor 2014, 50–55). Campbell notes that many of the "West Germanic" shared features could alternatively be treated as parallel developments rather than common inheritance (Campbell 1959, sects 1–5).

CAPR uses `PWGmc` and `NWGmc` as organizing labels for this chapter without claiming to have settled all questions about West Germanic subgrouping. Changes that appear in the literature under "Ingvaeonic" labels but affect the Old English–to-Proto-Germanic derivation chain are treated here as late expressions of the same West Germanic developmental period unless existing CAPR dossier research specifically argues for Anglo-Frisian or English-specific placement.

## 2.5 Rule names

The CAPR rules in this chapter carry names beginning with `NWGmc` or `PWGmc`. These names are stable internal identifiers. A name beginning with `NWGmc` does not guarantee that the change is exclusive to Northwest Germanic, and a name beginning with `PWGmc` does not guarantee that it is absent from North Germanic. The historical analysis in each sound-change section takes priority over the name prefix.

# 3 Root-noun nominative **-z* loss

## 3.1 Historical discussion

The athematic consonant stems — the "root nouns" of the handbooks — attached the nominative-singular marker directly to a consonant-final root, and the Proto-Germanic outcome of that collision is genuinely uncertain. Ringe gives the consonant-stem nominative ending as zero, **-z*, or possibly **-s*, and states plainly that the distribution is unrecoverable for monosyllabic stems (Ringe 2017, 306, §4.3.4). His own paradigm tables carry the uncertainty into print: the nominative of 'foot' appears as "fōts? (fōs?)", while 'mouse' is plain **mūs*, its expected extra sibilant already absorbed by degemination (Ringe 2017, 149, 313). Ringe and Taylor repeat the same three-way agnosticism — the root-noun nominative "either ended in **-s* or **-z*, or was endingless" (Ringe and Taylor 2014, 28, §2.3.1).

The dictionary traditions encode this situation in different notations, and the differences are conventions of citation rather than competing claims of fact. Orel prints morphologically explicit nominatives with final *-z* across the whole class: **bōkz* 'book', **gansz* 'goose', **lūsz* 'louse' (Orel

2003, 52, 126, 252). Kroonen cites the same words as bare stems or endingless forms, **bōk-* and **gans-* (Kroonen 2013, 71–72, 168–69), and Kluge/Seebold print a third variant with voiceless *-s*, as in **bōks* 'Buch' (Kluge 2011, 158). Bammesberger shows why the marker is nonetheless real: for voiced-final stems the overt nominative **-z* is positively reconstructible — **burgz* (Gothic *baúrgs*), **frijōndz* (Gothic *frijonds*) — even though **fōt-z* is "phonotaktisch kaum denkbar", and in West Germanic the ending simply "fiel es ab" (Bammesberger 1990, 190–92, §8.2.3.1). CAPR retains Orel's morphologically explicit forms as its inputs precisely because they record the inflectional marker whose fate this rule describes.

The three focal words are only superficially parallel. The root-final *s* of 'louse' is itself an extension of **luw-* on the model of 'mouse' (Bammesberger 1990, 195, §8.3). For 'goose', Szemerényi's lengthening would give a Proto-Indo-European nominative **ǵʰanss* > **ǵʰān*, after which the Proto-Germanic nominative was rebuilt with a final voiced sibilant, **ganz*, reanalyzed within the paradigm (Bammesberger 1990, 196, §8.3; Kroonen 2013, 168–69). 'Book' preserves a plain obstruent-final root. What unites them is not a single phonetic history but membership in a paradigm class that generalized the endingless nominative.

The branch evidence dates and localizes that generalization. Gothic keeps its sibilant throughout the class (*baúrgs*, *nahts*, *reiks*) (Fulk 2018, 165–66, §§7.26–7.27). Old Norse redistributes the ending morphologically: masculine root nouns keep *-r* (*fótr*), feminines are endingless (*nótt*, *geit*), while the vocalic-stem feminines *kýr*, *sýr*, *ær* retain *-r* from **-z* with R-umlaut (Fulk 2018, 167, §7.28; Bammesberger 1990, 192–93). West Germanic alone is uniform: there was "no ending in PWGmc, as none of the daughters exhibits any" (Ringe and Taylor 2014, 118, §3.4). Fulk supplies the mechanism: Szemerényi's law removed the nominative sibilant after sonorant-final stems, and endinglessness then spread analogically through the class (Fulk 2018, 143, §7.2). The development is therefore best understood as a morphological generalization enacted differently in each branch — absolute in West Germanic, consonant- and gender-conditioned in North Germanic, absent in Gothic — rather than as one exceptionless sound law.

This change is distinct from the two later final-*z developments. It was complete before Proto-West Germanic, whereas the loss of **-z* in unstressed syllables (SC020 EAFFinalZDeletion) is itself a Proto-West Germanic change: polysyllabic consonant-stem nominatives such as **fadurz* 'father' — and, in this corpus, **frijōndz* 'friend', **melukz* 'milk', and **mēnōþz* 'month' — kept their ending into Proto-West Germanic and lost it there in an unstressed syllable (Ringe and Taylor 2014, 44–45, §3.1.1). The still later northern loss of **-z* in stressed monosyllables (SC097 MonosyllabicFinalZLoss) affects vowel-final monosyllables like **hwaz* and does not touch the consonant-final root nouns at all, whose ending was gone long before.

## 3.2 SC096. Root-noun nominative **-z* loss (`RootNounNomZLoss`)

```
define RootNounNomZLoss [{*z} -> 0 ||
    .#. [EnglishStarConsonant | EnglishPalatalConsonant]*
        EnglishStarVocalic+
        [EnglishStarConsonant | EnglishPalatalConsonant]+ _ .#.];
```

The rule deletes word-final *z after a consonant in a monosyllable. Three claims of different kinds meet here and must be kept apart. The historical claim is morphological: the nominative-singular ending was lost in the athematic root-noun class, so that this one paradigm cell came to lack its marker — a development complete before Proto-West Germanic (Ringe and Taylor 2014, 118, §3.4). The lexical claim belongs to the dictionaries: Orel's citation forms, which supply the corpus inputs, print that marker explicitly as *-z* in **bōkz*, **flauxz*, **gansz*, and **lūsz* (Orel 2003, 52, 105, 126, 252). The executable statement is neither of these but a computational proxy for them: 'delete word-final *z after a consonant in a monosyllable'. It is not proposed as a Proto-Germanic sound law; it earns its place only because every form the corpus submits to the morphological development is a consonant-final monosyllable, so the narrow phonological statement covers the class exactly. Four corpus derivations witness the rule, each yielding its expected Old English outcome: PGmc **bōkz* 'book' yields OE *bōc* 'book', **gánsz* 'goose' yields *gōs* 'goose', **lūsz* 'louse' yields *lūs* 'louse', and **fláuxz* 'flea' yields *flēah* 'flea'. Should the corpus ever acquire a consonant-final monosyllable in **-z* that is not a root-noun nominative, the proxy and the morphology would come apart, and the rule would need to be re-scoped; the project's regression tests pin the firing population to exactly these four words so that any fifth firing forces that adjudication rather than passing silently.

The rule applies at the head of the English line, before SC009 PWGmcIjContraction. That ordering is fixed by the identity of the process rather than by a wrong form: contraction turns the polysyllabic **fríjōndz* 'friend' into a monosyllable, and if the root-noun rule applied after contraction it would capture **friundz* — yet the ending of 'friend' survived into Proto-West Germanic and fell in an unstressed syllable, the change described under SC020 EAFFinalZDeletion (Ringe and Taylor 2014, 44–45, §3.1.1). Because both rules delete the same segment, moving this rule later changes no Old English output; the early placement keeps the derivation of 'friend' aligned with the historical account rather than with an accident of the cascade.

Negative controls behave as the morphology predicts. Stressed monosyllables whose **z* follows a vowel — the domain of the later northern change — do not meet the post-consonantal environment. Medial **z* is untouched and remains available for rhotacism (SC003 EAFRhotacism): PGmc **déuzą* 'deer' still yields OE *dēor* 'deer' with its rhotacized medial consonant.

# 4 Early unstressed vowel changes

## 4.1 Historical discussion

The first change monophthongizes unstressed **ai*; the second carries early unstressed front-vowel leveling farther in forms such as *weorold* 'world'. Both have a diagnostic later boundary in the dataset.

## 4.2 Historical discussion of unstressed **ai* monophthongization

Ringe and Taylor describe the broad Northwest Germanic reduction of unstressed **ai* to a long mid vowel that merges with unstressed **e*, in final and nonfinal syllables alike (Ringe and Taylor 2014, 37–41). Two dative-singular endings in the dataset, span **spánnai* 'span' and meed **mízdai* 'meed', carry the change. The stressed development of **ái* to **ā* is treated separately as SC004 EAFAiMonophthongization.

## 4.3 SC014. Monophthongization of unstressed **ai* (PNWGmcUnstressedAiMonophthongization)

```
define PNWGmcUnstressedAiMonophthongization [
    {*ai} -> {*ē}
];
```

The dative-singular endings span **spánnai* 'span' and meed **mízdai* 'meed' carry this change; both give a final **ē*. If SC014 PNWGmcUnstressedAiMonophthongization is delayed until after SC072 OEUnstressedLongVowelShortening, the **ē* is no longer present for shortening, so PGmc **spánnai* 'span' yields †*spannē* rather than expected OE *spanne* 'span'. This shows that SC014 PNWGmcUnstressedAiMonophthongization must come before SC072 OEUnstressedLongVowelShortening in the modeled sequence.

Ringe and Taylor's merger of unstressed **ai* with long mid **ē* establishes the historical development, in final and nonfinal syllables alike. The stressed development of **ái* to **ā* is a separate and later change; see SC004 EAFAiMonophthongization.

## 4.4 Historical discussion of early unstressed front-vowel leveling

Campbell treats the merger of unstressed front vowels directly and also records the variation of *weorold* 'world' and *weoruld* 'world' (Campbell 1959, 141–42, 154–55). These forms supply SC015 PNWGmcILowering with a firmer lexical basis than the preceding change.

## 4.5 SC015. Leveling of early unstressed front vowels (PNWGmcILowering)

```
define PNWGmcILowering [
    {*i} -> {*e}
        || .#. EnglishStarNonVelarConsonant* _
           EnglishStarCoronal+ EnglishStarNonHighVowel,
    {*í} -> {*é}
        || .#. EnglishStarNonVelarConsonant* _
           EnglishStarCoronal+ EnglishStarNonHighVowel
];
```

The *weorold* 'world' and *weoruld* 'world' variants turn the general source claim into an ordering test. If SC015 PNWGmcILowering is delayed until after SC036 OEInterStressRaising, PGmc **wíràldu* 'world' yields †*wuruld* rather than expected OE *weorold* 'world'; earlier movement changes no output.

The derivation thus fixes front-vowel leveling before interstress raising while leaving its earlier boundary open.

SC016 OEWsPalatalGlide and SC017 PNWGmcULowering follow with a more tightly constrained local chronology.

# 5 Unstressed **a*-raising before final **m*

## 5.1 Historical discussion

Campbell notes that unstressed *u* is especially well preserved before *m*, with dat.pl. *-um* and related endings as the clearest evidence (Campbell 1959, 156, §373). Fulk likewise includes the development of early unstressed **o* to *u* before *m* among the similarities shared by North and West Germanic (Fulk 2018, 16, §5.2).

I restrict the change to unstressed vowels in inflectional material because the strongest evidence concerns noninitial unstressed material before final **m*. Final **m* conditions the raising.

## 5.2 SC005. Unstressed **a*-raising before final **m* (`PNWGmcAToUBeforeM`)

```
define PNWGmcAToUBeforeM [
    {*a} -> {*u} || EnglishStarVocalic EnglishStarConsonant+ _ {*m} ({*i})?
     ↪  ({*z})? .#.
];
```

Here the witness word and the comparative evidence serve different purposes. If raising is delayed until after SC017 PNWGmcULowering, PGmc **skúldramiz* ‘shoulders’ yields †*sċoldrum* rather than expected OE *sċuldrum* ‘shoulders’; earlier placements converge on the expected output. The scope of the change is established by inflectional evidence across multiple paradigm types: a-stem dative plural ON *dǫgum* ‘days’, OE *dagum* ‘days’, OS *dagun* ‘days’, OHG *tagum* ‘days’, beside Gothic *dagam* ‘days’; strong-adjective dative singular ON *góðum* ‘good’, OE *gōdum* ‘good’, OS *gōdum* ‘good’, beside Gothic *godamma* ‘good’ (OS also shows variant forms gōdumu and -un); and first-plural present ON *berum* ‘we carry’, OHG *berumēs* ‘we carry’, beside Gothic *baíram* ‘we carry’. Across these sets, North/West Germanic shows unstressed *-um* where Gothic preserves *-am*. The derivation of *sċuldrum* ‘shoulders’ supplies a CAPR ordering witness for the relative chronology, but the cognate set for ‘shoulder’ does not contribute comparative evidence for the rule’s historical scope.

# 6 Early i-apocope

## 6.1 Historical discussion

Sievers/Brunner treats the early loss of final **i* after unstressed syllables as established by the fact that these endings no longer trigger later i-umlaut in Old English, and Ringe and Taylor make the same point through the pathway to *geoguþ* 'youth' (Brunner 1965, sects 145–146; Ringe and Taylor 2014, 141). Campbell's *dugup* 'troop' and *geogup* 'youth' examples belong to the same pattern (Campbell 1959, sect. 332).

The ending vowel disappears in a weak suffixal environment early enough to block later umlaut. This anti-umlaut chronology distinguishes the change from later final-vowel losses.

## 6.2 SC006. Early i-apocope (`PWGmcEarlyIApocope`)

```
define PWGmcEarlyIApocope [
    {*i} -> 0 || PGmcStarStressedVowel PGmcStarConsonant+ PGmcStarVocalic
     ↪  PGmcStarConsonant+ _ .#.,
    {*i} -> 0 || PGmcStarStressedVowel PGmcStarConsonant+ PGmcStarVocalic
     ↪  PGmcStarConsonant+ _ {*z} .#.
];
```

The absence of umlaut in *geoguþ* 'youth' provides the historical argument for early deletion. The ordered derivation supplies a different test: if apocope is delayed until after SC034 OEAwLong-Diphthong, PGmc **skáwōθi* 'shows' yields †*sċēaweþ* rather than expected OE *sċēawaþ* 'shows'.

Early i-apocope must therefore precede the long-diphthong development. Moving it earlier within the tested range leaves every output unchanged; its early date rests on the anti-umlaut evidence, not on a lower boundary supplied by the witness words.

# 7 Final **ō*-lowering before **r*

## 7.1 Historical discussion

Ringe and Taylor separate two points here: a broader shortening of vowels before word-final **r* in unstressed syllables (for which kinship **r*-stems such as PGmc **fadér* > PWGmc **fader* are a key diagnostic), and the specific **ō*-before-**r* development needed here (Ringe and Taylor 2014, 58–59). The direct lexical witnesses for **ō* in that environment are two independent etyma: PGmc **fedwōr* 'four' with WGmc reflexes OE *fēower* 'four', OFris *fiuwer* 'four', OS *fiuwar* 'four'; and PGmc **watōr* 'water' with OE *wæter* 'water'.

The rule is historically secure but narrow: final or pre-final **ō* before word-final **r*. The clearest evidence remains concentrated in the `four` and `water` material. No broader environment for **ō* is attested.

## 7.2 SC007. Lowering of final bimoric **ō* before **r* (PWGmcFinalOrLowering)

```
define PWGmcFinalOrLowering [
    {*ō} -> {*a} || _ {*r} .#.
];
```

OE *wæter* 'water' reveals why lowering must precede SC043 EAFBrightening. If SC007 PWGmcFinalOrLowering is delayed until afterwards, PGmc **wátōr* 'water' yields †*water* rather than expected OE *wæter* 'water': brightening can affect the vowel only after lowering has created its input. Moving the change earlier within the tested range alters no output.

The witness thus supplies a terminus ante quem at brightening but no earlier boundary. Comparative support for the **ō*-before-**r* rule comes from the two lexical witnesses **fedwōr* 'four' and **watōr* 'water' and their WGmc reflexes; kinship **r*-stems belong to the broader pre-**r* shortening context, not to a direct **ō* > **a* control. Within CAPR, *wæter* 'water' is the form that establishes ordering before brightening. No broader lowering of **ō* is attested.

# 8 Coronal-w assimilation

## 8.1 Historical discussion

Ringe and Taylor treat the assimilation of **dw* and **zw* to **ww* as a shared Proto-West-Germanic innovation supported by one example of each input cluster (Ringe and Taylor 2014, 56–57; Stiles 1985, 89–94). The **dw* example is the numeral 'four': PGmc **feðwor* (Gothic *fidwor*) → WGmc **fewwar* → OE *fēower*, Old Frisian *fiuwer*, Old Saxon *fiuwar*. The **zw* example is the second-person plural pronoun, where two oblique case forms show the change: acc./dat. PGmc **izwiz* (Gothic *izwis*) → OE *eow*, Old Frisian *iu*, Old Saxon *iu*, OHG *iu*; and gen. Ringe and Taylor's PGmc **izweraz* (Gothic *izwara*) → OE *eower*, OHG *iuwer* (Ringe and Taylor 2014, 56). Stiles discusses the same pronominal material using his own reconstruction conventions and explicitly treats Gothic *izwara* among the relevant comparanda (Stiles 1985, 89–94). These two case forms belong to a single pronominal paradigm, not to two independent etyma.

The historical support rests on a small witness set. Both coronal inputs assimilate before **w*, but the evidence for each cluster is confined: the numeral alone supplies the **dw* instance, and the oblique case forms of the second-person plural pronoun supply the **zw* instance. Both clusters are now witnessed in the corpus: 'four' for **dw*, and 'you' — selected in its dat.(-acc.) plural cell **izwiz* — for **zw*, deriving through **iwwi*, apocopated **iww*, to OE *ēow* 'you' (Ringe and Taylor 2014, 41–42, §3.1.1; Fulk 2018, sect. 8.3, pp. 204–205).

## 8.2 SC008. Assimilation of coronal consonants before **w* (PWGmcCoronalWAssimilation)

```
define PWGmcCoronalWAssimilation [
    {*d} -> {*w} || _ {*w},
    {*z} -> {*w} || _ {*w}
];
```

OE *fēower* 'four' exposes a feeding relation: coronal assimilation must create **ww* while simplification can still reduce it. If SC008 PWGmcCoronalWAssimilation is delayed until after SC031 OEWWSimplification, PGmc **fédwōr* 'four' yields †*fēowwer* rather than expected OE *fēower* 'four'. Earlier placements alter no output.

The numeral fixes that relative order. The pronoun now fixes a second one: assimilation must precede rhotacism (SC003 EAFRhotacism). The **z* of **izwiz* stands between vowel and **w*; had rhotacism applied first, it would have produced †*irwiz*, from which OE *ēow* 'you' can never be derived. The executable cascade composes the assimilation well before rhotacism, and the corpus derivation of *ēow* 'you' fails if the two are reversed. 'Four' remains the sole **dw* witness and the sole source of the coronal-assimilation → *ww*-simplification ordering constraint. The earlier boundary of the assimilation remains undetermined.

# 9 *ij*-contraction in *friend*

## 9.1 Historical discussion

Ringe and Taylor describe a change of **ijo* to **iu* in the ancestor of *friend*, with the pathway PGmc **frijōnd-* (Gothic *frijonds*) $\longrightarrow$ PWGmc **friund* $\longrightarrow$ OE *frēond*, Old Frisian *frīund*, Old Saxon *friund*, Old High German *friunt* (Ringe and Taylor 2014, 62). The same source immediately warns that the **ijo* sequence is unique enough that wider generalization is inadvisable (Ringe and Taylor 2014, 62). Luick (printed p. 118) notes that *iu* generalised within several *j*-stem paradigms through a related but differently conditioned loss of *j*, but does not supply a second example of the exact stressed **ijo* sequence (Luick 1914--1940, 118).

The change concerns a rare sequence attested only in the **frijōnd-* etymon and cannot safely be generalized into a broadly productive rule.

## 9.2 SC009. *ij*-contraction in *friend* (`PWGmcIjContraction`)

```
define PWGmcIjContraction [
    {*i} {*j} {*ō} -> {*iu} || _ EnglishStarConsonant,
    {*í} {*j} {*ō} -> {*íu} || _ EnglishStarConsonant
];
```

Only the **frijōnd-* etymon tests this contraction. If the rare **ijō* sequence survives until after SC032 OEDiphthongLeveling, PGmc **fríjōndz* 'friend' yields †*friund* rather than expected OE *frēond* 'friend'; moving contraction earlier within the tested range changes no output.

That single contrast places SC009 PWGmcIjContraction before diphthong leveling but gives no lower boundary. It cannot establish a productive sound law beyond the **frijōnd-* etymon, precisely the reservation made by Ringe and Taylor.

# 10 West Germanic j-gemination

## 10.1 Historical discussion

Fulk treats West Germanic consonant gemination before `*j` after a short vowel as a regular development and illustrates it with forms such as OE *settan* ‘set’ and *lecgan* ‘lay’ (Fulk 2018, 127, §6.15).

The change applies specifically after a short vowel before **j*, not to geminate consonants generally.

## 10.2 SC010. West Germanic j-gemination (`PWGmcJGemination`)

```
define PWGmcJGemination [
    {*p} -> {*p} {*p} || EnglishStarShortVowel _ {*j},
    {*b} -> {*b} {*b} || EnglishStarShortVowel _ {*j},
    {*t} -> {*t} {*t} || EnglishStarShortVowel _ {*j},
    {*d} -> {*d} {*d} || EnglishStarShortVowel _ {*j},
    {*k} -> {*k} {*k} || EnglishStarShortVowel _ {*j},
    {*g} -> {*g} {*g} || EnglishStarShortVowel _ {*j},
    {*f} -> {*f} {*f} || EnglishStarShortVowel _ {*j},
    {*s} -> {*s} {*s} || EnglishStarShortVowel _ {*j},
    {*m} -> {*m} {*m} || EnglishStarShortVowel _ {*j},
    {*n} -> {*n} {*n} || EnglishStarShortVowel _ {*j},
    {*l} -> {*l} {*l} || EnglishStarShortVowel _ {*j},
    {*ŋ} -> {*ŋ} {*ŋ} || EnglishStarShortVowel _ {*j},
    {*x} -> {*x} {*x} || EnglishStarShortVowel _ {*j}
];
```

OE *nett* ‘net’ fixes the order because the syllabic-*j* development would remove the glide that conditions gemination. If SC011 PWGmcSyllabicJ precedes SC010 PWGmcJGemination, PGmc **nátją* ‘net’ yields †*nete* rather than expected OE *nett* ‘net’. Earlier movement of gemination changes no output.

The chronology is phonologically transparent: the consonant must geminate before **j* ceases to be consonantal. The witness establishes no earlier boundary.

# 11 Syllabic j after final-vowel loss

## 11.1 Historical discussion

Ringe and Taylor state directly that after final unstressed `*a` and `*ą` were lost, postconsonantal `*j` became syllabic `*i`, with outcomes behind OE *here* 'army' and *rice* 'kingdom' (Ringe and Taylor 2014, 46).

The sources establish the development, although the lexical evidence supplies little independent support for its position. Its scope is postconsonantal j, not high-vowel vocalization generally.

## 11.2 SC011. Syllabic **j* after final-vowel loss (`PWGmcSyllabicJ`)

```
define PWGmcSyllabicJ [
    {*j} {*a} -> {*i} || EnglishStarShortVowel EnglishStarConsonant _ .#.,
    {*j} {*ą} -> {*i} || EnglishStarShortVowel EnglishStarConsonant _ .#.
];
```

The same PGmc **nátją* 'net' witness supplies the only firm boundary. Placing SC011 PWGmcSyllabicJ before SC010 PWGmcJGemination yields †*nete* rather than expected OE *nett* 'net'; moving it later changes no output.

Comparative evidence establishes postconsonantal **j* to syllabic **i* after final unstressed **a* or **ą* loss, with *here* 'army' and *rice* 'kingdom' as outcomes. The lexicon adds only that vocalization followed gemination, not where it falls among subsequent changes.

# 12 *lþ*-voicing

## 12.1 Historical discussion

Ringe and Taylor treat word-internal **lþ* > **ld* as a regular sound change in northern West Germanic and illustrate it with forms such as *fealdan* 'fold', *beald* 'bold', *wuldor* 'glory', and *gylden* 'golden' (Ringe and Taylor 2014, 170–71). Campbell gives a similar West-Germanic-facing formulation with examples such as *fealdan*, *wuldor*, *beald*, *gold* 'gold', and *feld* 'field' (Campbell 1959, 169, §414).

The comparative evidence supports *lþ* > *ld* most clearly in northern West Germanic, not as an unqualified pan-PWGmc development.

## 12.2 SC012. Northern West Germanic *lþ*-voicing (`EAFLThVoicing`)

```
define EAFLThVoicing [
    {*θ} -> {*d} || {*l} _
];
```

The `field`, `fold`, `gold`, and `wold` families preserve **lþ* to **ld*, but none dates the change against a neighboring rule. Every output remains unchanged when the voicing is moved in either direction.

Comparative reconstruction therefore establishes northern West Germanic *lþ* > *ld*, but the witness forms fix no date. Neither a pan-PWGmc attribution nor an exact local placement follows from the evidence presented here.

# 13 Dental hardening

## 13.1 Historical discussion

Ringe and Taylor state directly that in PWGmc voiced dental fricative `*ð` became stop `*d` in all positions (Ringe and Taylor 2014, 43).

The change is systemic across early West Germanic and extends beyond any one lexical family.

## 13.2 SC013. Dental hardening (`PWGmcDentalHardening`)

```
define PWGmcDentalHardening [
    {*ð} -> {*d}
];
```

Dental hardening has systemic scope: voiced fricative **ð* became stop **d* throughout early West Germanic. Moving SC013 PWGmcDentalHardening earlier or later changes no output.

Comparative evidence establishes the sound law; the present lexicon leaves its exact position approximate.

# 14 West Saxon palatal glide before back vowels

## 14.1 Historical discussion

West Saxon spellings such as *ġeoc* ‘yoke’, *ġeong* ‘young’, and *ġeoguþ* ‘youth’ reflect an early Old English development before back vowels. Campbell gives the most direct handbook statement of the phenomenon (Campbell 1959, 17, §44).

The sources establish SC016 OEWsPalatalGlide, although the lexical evidence establishes only a later boundary. The rule is computationally positioned before SC017 PNWGmcULowering because the derivation of *ġeoc* ‘yoke’ requires glide insertion before u-lowering applies. That computational dependency places SC016 OEWsPalatalGlide in the Old English section of the cascade even though the cascade position precedes many Northwest Germanic changes.

## 14.2 SC016. West Saxon palatal glide before back vowels (OEWsPalatalGlide)

```
define OEWsPalatalGlide [
    {*j} {*u} -> {*j} {*e} {*u} || .#. _,
    {*j} {*ú} -> {*j} {*é} {*u} || .#. _
] .o. [
    {*ʤ} {*u} -> {*ʤ} {*e} {*u} || .#. _,
    {*ʤ} {*ú} -> {*ʤ} {*é} {*u} || .#. _
] .o. [
    {*ʧ} {*u} -> {*ʧ} {*e} {*u} || .#. _,
    {*ʧ} {*ú} -> {*ʧ} {*é} {*u} || .#. _
] .o. [
    {*ʃ} {*u} -> {*ʃ} {*e} {*u} || .#. _,
    {*ʃ} {*ú} -> {*ʃ} {*é} {*u} || .#. _
];
```

OE *ġeoc* ‘yoke’ fixes the close relation between glide insertion before back-vocalic *u* and the following change.

If glide insertion follows SC017 PNWGmcULowering, PGmc **júką* ‘yoke’ yields †*ġoc* rather than expected OE *ġeoc* ‘yoke’; earlier placement changes no output. The witness therefore dates SC016 OEWsPalatalGlide before u-lowering without supplying an earlier boundary. The *ġeoc* ‘yoke’, *ġeong* ‘young’, and *ġeoguþ* ‘youth’ material establishes the lexical scope of the West Saxon development.

# 15 Northwest Germanic u-lowering

## 15.1 Historical discussion

The derivation of *ġeoc* ‘yoke’ passes through both this change and the preceding West Saxon palatal glide (SC016 OEWsPalatalGlide). Campbell treats the West Saxon rising-diphthong spellings before back vowels, while the same handbook tradition describes the lowering of *u* before a following non-high vowel separately (Campbell 1959, 17, §44; Campbell 1959, 42–43, §115; Fulk 2018, 56, §4.3). The first change creates the West Saxon *ġeoc* type; u-lowering then carries the same material into the subsequent vowel history.

After the glide-conditioned West Saxon spellings are in place, the broader Northwest Germanic lowering of *u* to *o* before a following non-high vowel provides the clearest standard sound change in this small region. Campbell and Fulk both describe that change directly (Campbell 1959, 42–43, §115; Fulk 2018, 56, §4.3).

SC017 PNWGmcULowering thus rests on a broader source base than the preceding West Saxon rule.

## 15.2 SC017. Lowering of ***u*** before following non-high vowels (PNWGmcULowering)

```
define PNWGmcULowering [
    {*u} -> {*o}
        || .#. EnglishStarConsonant* _
           [EnglishStarConsonantNoJ - EnglishStarNasal]
           EnglishStarConsonantNoJ* EnglishStarNonHighVowel,
    {*ú} -> {*ó}
        || .#. EnglishStarConsonant* _
           [EnglishStarConsonantNoJ - EnglishStarNasal]
           EnglishStarConsonantNoJ* EnglishStarNonHighVowel
];
```

Lowering of *u* to *o* is fixed on both sides by *ġeoc* ‘yoke’, *nosu* ‘nose’, *sċofl* ‘shovel’, and *sorg* ‘sorrow’.

Before SC016 OEWsPalatalGlide, PGmc **júką* ‘yoke’ yields †*ġoc* rather than expected OE *ġeoc* ‘yoke’. After SC019 PNWGmcFinalLongORaising, PGmc **núsō* ‘nose’ yields †*nusu* rather than expected *nosu* ‘nose’, PGmc **skúflō* ‘shovel’ yields †*sċufl* rather than expected *sċofl* ‘shovel’, and PGmc **súrgō* ‘sorrow’ yields †*surg* rather than expected *sorg* ‘sorrow’. The two witness sets place SC017 PNWGmcULowering after glide formation and before final long-*o* raising.

# 16 Stressed monosyllable *ō-raising

## 16.1 Historical discussion

Campbell treats the development of final accented *ō* to *ū* in stressed monosyllables directly, with the familiar outcomes behind *cū* 'cow', *hū* 'how', *tū* 'two', and *bū* 'both' (Campbell 1959, 47, §122).

The change is historically secure, but the tested forms determine no close relative position for it. Its input is final **ō* in a stressed monosyllable.

## 16.2 SC018. Raising of final stressed monosyllabic *ō (PNWGmcStressedMonosyllableORaising)

```
define PNWGmcStressedMonosyllableORaising [
    {*ō} -> {*ū} || .#. [EnglishStarConsonant | EnglishPalatalConsonant]* _ .#.
];
```

Campbell's *cū* 'cow', *hū* 'how', and *tū* 'two' establish final stressed monosyllabic **ō* > **ū*.

Reversing SC018 PNWGmcStressedMonosyllableORaising with neighboring changes leaves every output unchanged. The sound change is secure, but its exact position in the early history of long vowels rests on the handbooks.

# 17 Raising of final unstressed long **ō*

## 17.1 Historical discussion

Ringe and Taylor describe the change of unstressed final non-nasalized long **ō* to short **u* as a Northwest Germanic development (Ringe and Taylor 2014, 30). It applies in the same final-syllable environment as the subsequent loss of word-final **z*. The derivation of *ræste* 'rest' fixes their local order: SC019 PNWGmcFinalLongORaising must still see final **ō*, and word-final **z*-deletion removes the following **z* only afterward.

The change supplies the final vowel of forms such as *nosu* 'nose', *sċofl* 'shovel', and *sorg* 'sorrow'.

## 17.2 SC019. Raising of final unstressed long **ō* (PNWGmcFinalLongORaising)

```
define PNWGmcFinalLongORaising [
    {*ō} -> {*u}
        || EnglishStarVocalic
           [EnglishStarConsonant | EnglishPalatalConsonant]+ _ .#.
];
```

Two groups of witnesses confine final unstressed long **ō* > **u*. The forms *nosu* 'nose', *sċofl* 'shovel', and *sorg* 'sorrow' fix its lower boundary.

Before SC017 PNWGmcULowering, PGmc **núsō* 'nose' yields †*nusu* rather than expected OE *nosu* 'nose', PGmc **skúflō* 'shovel' yields †*sċufl* rather than expected *sċofl* 'shovel', and PGmc **súrgō* 'sorrow' yields †*surg* rather than expected *sorg* 'sorrow'. After word-final **z*-deletion (SC020 EAFFinalZDeletion), PGmc **rástōz* 'rest' yields †*rast* rather than expected *ræste* 'rest'. These failures place SC019 PNWGmcFinalLongORaising after u-lowering and before final *z*-loss.

# 18 Chapter 2. From Proto-West Germanic to Anglo-Frisian

## 18.1 Historical interval

This chapter covers the sound changes that occurred after the Proto-West Germanic period and before, or during the emergence of, the specifically English line. The starting reconstruction is Proto-West Germanic; the end point is the Proto-Anglo-Frisian stage — or more precisely, the cluster of innovations that define the English and Frisian branch within West Germanic.

## 18.2 A necessary terminological caution

The title of this chapter uses "Anglo-Frisian" as an organizing historical concept. That choice requires an explicit qualification.

The scholarly literature uses several overlapping terms for this developmental period:

- North Sea Germanic and Ingvaeonic: labels used by some scholars for a proposed subgroup comprising Old English, Old Frisian, and Old Saxon (or more narrowly, Old English and Old Frisian only). The innovations associated with this label — especially the nasal spirant changes and certain vowel developments — are sometimes described as diffusion rather than shared inheritance (Campbell 1959, sects 1–3).
- Anglo-Frisian: a label used specifically for the Old English / Old Frisian branch, or for innovations shared between the two languages. Its use presupposes a tighter relationship between English and Frisian than between either and Old Saxon.
- Proto-Anglo-Frisian (PAF): the strongest interpretation, positing a discrete reconstructed common ancestor for Old English and Old Frisian specifically. This is the position of Ringe and Taylor, who reconstruct a PAF stage between Proto-West Germanic and Proto-Old-English (Ringe and Taylor 2014, 54–68).

CAPR does not commit to a universally accepted discrete PAF node. The chapter title uses "Anglo-Frisian" because the key changes of this period — especially West Germanic rhotacism (SC003), word-final `*z` deletion (SC020), and Anglo-Frisian brightening (SC043) — are most prominently associated with that label in the handbook literature. But the analysis does not require that every change passed through a single genealogical PAF stage. Some changes may be West Germanic broadly; others may reflect areal diffusion. The existing CAPR dossiers record the source-by-source picture where these distinctions matter.

## 18.3 Major changes and their historical basis

### 18.3.1 West Germanic rhotacism (SC003)

The medial change of `*z` to `*r` in environments such as **déuzaz* 'deer', **xúrdaz* 'hoard', and **líznōjaną* 'learn' is historically a post-Proto-West-Germanic development. Ringe and Taylor argue that rhotacism was not inherited from Proto-Northwest Germanic and was not uniform within West Germanic (Ringe and Taylor 2014, 52, 98, 102). Crist separates this change explicitly from the deletion of word-final `*z` and argues that rhotacism must follow the deletion rules (Crist 2001, 104–6; 2002, 1, 4). Hogg gives the standard Old English–facing summary: `*z` yielded `*r` in intervocalic position but was generally lost in final position (Hogg 1992, 37).

The CAPR rule is named `EAFRhotacism`, placing it in the Early Anglo-Frisian corridor, CAPR's operational post-Proto-West-Germanic stage on the English line; the reader-facing chapter label describes the change as a West Germanic rhotacism.

### 18.3.2 Word-final *z deletion (SC020) and the three final-*z developments

The loss of word-final `*z` is not one process but three historically distinct developments, and this chapter contains two of them. The central one, SC020, is the Proto-West Germanic loss of `*z` in unstressed syllables, seen in forms such as **rástōz* 'rest (nom.sg.)' and stated for the whole branch by Ringe and Taylor (Ringe and Taylor 2014, 44–45); Crist's analysis distinguishes it both from the earlier NWGmc changes and from the later narrower Ingvaeonic deletion rules (Crist

2002, 1, 4). Earlier still, the consonant-stem root nouns had generalized endingless nominatives before Proto-West Germanic (SC096, Chapter 1). Later, and only in the north, `*z` was lost in stressed monosyllables with compensatory lengthening (SC097, this chapter); the southern dialects instead retained and rhotacized it. The standard handbooks confirm the West Germanic deletion in general terms: Campbell notes that `*z` is "later lost or changed to `r`" (Campbell 1959); Hogg gives a clean statement that Germanic `*z` is generally lost in final position (Hogg 1992, 37); the three-way division refines those summaries rather than contradicting them.

The CAPR rule for the unstressed loss is still named `EAFFinalZDeletion`, an identifier that predates the restaging of the change to Proto-West Germanic; the name is retained as a stable identifier pending a global renaming pass, and the historical stage recorded in the staging metadata takes priority over the name prefix. SC020 remains presented in this chapter, beside rhotacism, because the two changes jointly determine the fate of every remaining `*z`.

### 18.3.3 Anglo-Frisian ai-monophthongization (SC004)

The monophthongization of stressed/root `*ái` to `*ā`, seen in the soul derivation, is a North Sea Germanic areal development. Versloot argues that it spread in successive waves through a dialect continuum, with Old English among the widest to carry it, so the change is better read as an areal diffusion through the continuum than as a single dated node (Versloot 2017, 281–324). The resulting `*ā` is later fronted to ǣ in the relevant Old English environments.

The CAPR rule is named `EAFAiMonophthongization` and executes at cascade position 28. That position is CAPR's operational home for a North Sea Germanic areal change on the English line; it is a modelling choice, not a claim that the change passed through a discrete Proto-Anglo-Frisian node. The one usable chronological anchor is the `soul` derivation, which requires the monophthongization to precede OE interstress raising (SC036).

The unstressed development `*ai > *ē` (in final and nonfinal syllables) is a separate and earlier Proto-Northwest Germanic change (SC014), discussed in Chapter 1; its corpus witnesses are the dative-singular endings of `span` (**spánnai* 'span' > *spanne* 'span') and `meed` (**mízdai* 'meed' > *meorde* 'meed').

### 18.3.4 Anglo-Frisian brightening (SC043, treated in Chapter 3)

The fronting of low `*a` to `*æ` outside nasal environments is the defining Anglo-Frisian change. It executes later in the cascade than the changes of this chapter, so its full section appears in Chapter 3; it is introduced here because it anchors the "Anglo-Frisian" label that names this period. Campbell gives the classical statement: "By a very early change Prim. Gmc. `a > æ` in OE and OFris. when not followed by a nasal consonant" (Campbell 1959, sects 163–165). Hogg gives the most familiar modern label pair: "This vowel normally fronted to /ae/ by the sound change of Anglo-Frisian Brightening (or First Fronting)" (Hogg 1992, sect. 5.8).

The change is notable for what follows it: OE Breaking presupposes the fronted input; OE a-Restoration partially undoes it in back-vowel environments. The three-change sequence (brightening, breaking, restoration) is one of the clearest relative-chronology chains in the Old English historical grammar.

Campbell notes that English and Frisian may not simply reflect one undifferentiated shared prehistoric event, and Ringe and Taylor leave open whether the wider spread of fronted outcomes happened mainly on the continent or in Britain (Ringe and Taylor 2014, 60–62). CAPR's implementation treats the change as a single rule; the book prose acknowledges the uncertainty about its exact geographical scope.

The current CAPR inventory has this change labeled "Old English" in the pipeline taxonomy (no separate Anglo-Frisian bucket previously existed). The historical staging map places its section in Chapter 3, at its executable cascade position.

## 18.4 Cascade vs. historical order in this chapter

The changes in this chapter are presented in the order in which they apply in the executable cascade, which models the reconstructed chronology itself: first word-final `*z` deletion in unstressed syllables (SC020), then the later northern monosyllabic `*z`-loss (SC097) that completes the final-`*z` story, then West Germanic rhotacism (SC003), which turns every surviving `*z` into `*r` only after the deletions have run their course. There follow the unstressed `*ō` raising (SC021), the `*mn` dissimilation and n-stem `*n` loss (SC022–SC023), the long-`*ē` developments (SC024–SC025), the nasal-spirant corridor (SC026–SC027) with preconsonantal `*x` loss (SC028), and finally Anglo-Frisian ai-monophthongization (SC004, the North Sea areal vowel change) closing the chapter.

Book order, cascade order, and reconstructed historical order coincide here: in particular, the deletions of final `*z` precede rhotacism both in the sources and in the executable derivation, so no form ever meets rhotacism with a word-final sibilant intact.

# 19 West Germanic final **z*-deletion

## 19.1 Historical discussion

Word-final **z* in unstressed syllables was lost in Proto-West Germanic. Ringe and Taylor state the change for the whole branch and illustrate it with the nominative plural **dagōz* > **dagō* and the consonant-stem nominative **fadurz* > **fadur*, noting that the ending is lost after consonants as well as after vowels (Ringe and Taylor 2014, 44–45, §3.1.1). Crist's handout formulates the same development and its Ingvaeonic sequels (Crist 2002, 2, §§5–6). The change is pan-West-Germanic, not specifically Ingvaeonic: every West Germanic daughter shows the loss, and the Frienstedt comb inscription *kaba* < **kambaz* 'comb' (c. 250–300 CE) supplies early epigraphic confirmation (Fulk 2018, 25, n. 1).

The conditioning segment is specifically the voiced sibilant **z*, never **s*: Ringe and Taylor's near-minimal pair of nominative singular **dagaz* > **dag* beside genitive singular **dagas*, which keeps its sibilant into Old English *dæġes*, shows that the change reads the Verner voicing distinction (Ringe and Taylor 2014, 212, §6.1). Where the handbooks disagree about whether a given ending had **-s* or **-z* — as for the nominative plural **-ōz* — the disagreement matters directly to whether this rule applies (Ringe and Taylor 2014, 115–16, §4.2.1).

This is the middle of three historically distinct final-**z* developments, and Ringe and Taylor explicitly separate it from the later loss in stressed monosyllables, citing Crist's demonstration that they are two changes (Ringe and Taylor 2014, 44–45, §3.1.1). Earlier, the consonant-stem (root-noun) nominatives of monosyllables had already generalized endinglessness before Proto-West Germanic (SC096 RootNounNomZLoss), so forms like **bōkz* 'book' never reach this rule with their marker intact. Later, and only in the north, **z* was lost in stressed monosyllables with compensatory lengthening (SC097 MonosyllabicFinalZLoss); the present rule leaves stressed monosyllables untouched. Older accounts that grouped all of these under one loss of final **z*, such as Campbell's, are superseded by this three-way division (Campbell 1959, 166).

At the boundary with Chapter 1's Northwest Germanic sequence, the derivation of *ræste* 'rest' shows that final **ō*-raising (SC019 PNWGmcFinalLongORaising) must precede this rule: raising applies to **-ō* but not to **-ōz*, whose final vowel is still sheltered by the sibilant when raising runs (Ringe and Taylor 2014, 15–16, 24). On the later side, Ringe and Taylor order the loss of **z* before the loss of word-final bare **-a*, since **dagaz* first becomes **daga* and only then **dag* (Ringe and Taylor 2014, 45–46, §3.1.2).

## 19.2 SC020. West Germanic final **z*-deletion (`EAFFinalZDeletion`)

```
define EAFFinalZDeletion [{*z} -> 0 ||
    .#. ?* EnglishStarVocalic
        [EnglishStarConsonant | EnglishPalatalConsonant]+
        EnglishStarVocalic ?* _ .#.,
    .#. [EnglishStarConsonant | EnglishPalatalConsonant]*
```

```
        EnglishStarVocalic+
        [EnglishStarConsonant | EnglishPalatalConsonant]+ _ .#.];
```

The rule deletes word-final *z in unstressed syllables, stated through two environments. The first clause covers polysyllables, where the final syllable of these corpus forms is unstressed: this is the ordinary case, with 110 corpus derivations, such as PGmc **bárdaz* 'beard' on its way to OE *beard* 'beard' and **rástōz* 'rest' on its way to *ræste* 'rest'. The second clause covers post-consonantal *z in monosyllables. By the time this rule runs, SC096 RootNounNomZLoss has already removed the genuine root-noun nominative endings, so the only form reaching the second clause is **fríjōndz* 'friend', contracted to monosyllabic **fríundz* by SC009 PWGmcIjContraction; its ending, like that of **fadurz*, stood in an unstressed syllable when the Proto-West Germanic change applied and so belongs here rather than to the root-noun development (Ringe and Taylor 2014, 44–45, §3.1.1). Stressed monosyllables ending in vowel plus *z meet neither clause and are left for SC097 MonosyllabicFinalZLoss.

The chronology of word-final *z-loss is unusually well delimited: *ræste* 'rest' supplies its early boundary, while later weak syllables supply its late boundary.

Before SC019 PNWGmcFinalLongORaising, PGmc **rástōz* 'rest' yields †*rast* rather than expected OE *ræste* 'rest'. After SC040 OEMedUnstressedULowering, PGmc **bébruz* 'beaver' yields †*befro* rather than expected *befer* 'beaver', PGmc **kwéðuz* 'cud' yields †*cwedo* rather than expected *cwedu* 'cud', and PGmc **félθuz* 'field' yields †*feldo* rather than expected *feld* 'field', alongside eight other newly failing rows. Final *z*-loss therefore follows SC019 PNWGmcFinalLongORaising and precedes SC040 OEMedUnstressedULowering.

The **rástōz* 'rest' derivation fixes the local relation to SC019 PNWGmcFinalLongORaising. The distant boundary at SC040 OEMedUnstressedULowering shows only that word-final *z-loss precedes the later weak-syllable sequence; its placement within that wider interval follows the handbook chronology after final *ō-raising.

# 20 Early apocope in unstressed words

## 20.1 Historical discussion

Alongside the regular loss of word-final short high vowels in third syllables (SC006 PWGmcEarlyIApocope), Ringe and Taylor identify a second, earlier apocope: "Short high vowels were also lost after heavy syllables in unstressed words" (Ringe and Taylor 2014, 57–58, §3.1.4). The two laws must not be conflated. Fully stressed disyllables kept their final **-i* long enough to cause i-umlaut — OE *ġiest* 'guest' < **gastiz* and *fȳr* 'fire' < **fūri* require exactly that survival (Ringe and Taylor 2014, 55, §3.1.4) — whereas words that carried no sentence stress lost the vowel already in Proto-West Germanic.

The conditioning is prosodic. Like Verner's law, the change is governed by accent: it applied in words unstressed in the sentence, and its apparent exceptions are systematic, not sporadic. Forms such as OE *ymbe* 'around' and OHG *umbi* kept their final vowel because, as Ringe and Taylor observe, proclitics "were not phonologically word-final" and so stood outside the environment altogether (Ringe and Taylor 2014, 57–58, §3.1.4). Sentence-level accent placement therefore decides which sandhi variant each daughter language continues, and doublets across the family reflect the stressed and unstressed sentence forms of the same word — regular sandhi, not lexical diffusion.

The diagnostic witness is the second-person plural pronoun. Ringe and Taylor print the Proto-West Germanic form as a doublet: PGmc **izwiz* (Gothic *izwis*) $\rightarrow$ **iwwi* (by SC008 PWGmcCoronalWAssimilation and the loss of final *z, SC020 EAFFinalZDeletion) $\rightarrow$ PWGmc **iuwi* ~ **iuw* (Ringe and Taylor 2014, 41–42, §3.1.1). Old English continues the apocopated, unstressed variant, and Ringe and Taylor's proof is the vocalism itself: "OE iow 'you (dat. pl.)' definitely does [show early apocope] (since it does not exhibit i-umlaut)" (Ringe and Taylor 2014, 57–58, §3.1.4). Had the **-i* survived, i-umlaut (SC055 OEIUmlaut) would have fronted the diphthong; West Saxon *ēow* 'you' beside early West Saxon and Northumbrian *īow* shows the normal unumlauted development (Campbell 1959, sect. 702, p. 283).

## 20.2 SC098. Early apocope in unstressed words (PWGmcUnstressedWordFinalIApocope)

```
define PWGmcUnstressedWordFinalIApocope [
    {*i} -> 0 || {*w} {*w} _ .#.
];
```

The corpus transcription does not mark the absence of word stress, so the rule states the law through a proxy environment: word-final **-i* after the geminate **ww* created by coronal-w assimilation, which in the present corpus is exactly coextensive with the law's unstressed-word domain. The same convention serves SC096 RootNounNomZLoss, where a development whose true conditioning the notation cannot yet express is likewise implemented over an exactly coextensive segmental environment.

The corpus witness is 'you': **izwiz* $\rightarrow$ **iwwiz* (assimilation) $\rightarrow$ **iwwi* (final *z-loss) $\rightarrow$ **iww* (this rule) $\rightarrow$ OE *ēow* 'you'. The chronology is fixed on both sides. The rule is fed by the loss of final *z (SC020 EAFFinalZDeletion), since only that loss makes the **-i* word-final; and it must precede i-umlaut (SC055 OEIUmlaut) — that ordering is Ringe and Taylor's own dating argument, for an unapocopated **iwwi* surviving to the umlaut period would yield an umlauted diphthong and a form other than the attested *ēow* 'you'. The rule applies within Proto-West Germanic, before the later northern loss of final *z in stressed monosyllables (SC097 MonosyllabicFinalZLoss): Ringe and Taylor treat the apocope among the Proto-West Germanic final-syllable developments and print the apocopated variant as a Proto-West Germanic form (Ringe and Taylor 2014, 41–42, 57–58). Fully stressed disyllables are untouched, as the history requires: **gastiz* and **fūri* pass through unchanged and duly umlaut to *ġiest* 'guest' and *fȳr* 'fire'.

The surviving word-final geminate **ww* is then vocalized to a long diphthong (SC033 OEEwLongDiphthong) before geminate simplification (SC031 OEWWSimplification) can destroy it — Ringe and Taylor date that vocalization to Proto-West Germanic itself (**fewwar* → PWGmc **feuwar*) (Ringe and Taylor 2014, 41–42, §3.1.1; Fulk 2018, sect. 8.3, pp. 204–205) — giving **ēoww*, simplified to **ēow*, the attested Old English form.

# 21 Northern monosyllabic final *z-loss

## 21.1 Historical discussion

Long after the Proto-West Germanic loss of final *z in unstressed syllables (SC020 EAFFinalZDeletion), the northern West Germanic dialects lost word-final *z in stressed monosyllables as well, with compensatory lengthening of a short nucleus. Ringe and Taylor's witness set is OE *mā* 'more' < **maiz*, the pronouns *wē* 'we', *ġē* 'you', *mē* 'me', *þē* 'thee', *hē* 'he', *hwā* 'who' < **hwaz*, and — hedged in their own print with question marks — *cū* 'cow' (Ringe and Taylor 2014, 86, §3.3.1). The southern dialects retained the sibilant and rhotacized it: Old High German *mir*, *wir*, *mēr*, *er* answer the Old English endingless forms, which is why Fulk counts this loss among the diagnostic Ingvaeonic features (Fulk 2018, 18, n. 6). Ringe and Taylor explicitly treat this as a change separate from the Proto-West Germanic unstressed loss, citing Crist's demonstration that the two must be distinguished (Ringe and Taylor 2014, 44–45, §3.1.1).

The scholarship disagrees about the exact conditioning, and the disagreement is worth recording. An older account, represented by Campbell and going back to Luick, derived the endingless pronouns from unaccented sentence variants rather than from a regular sound change (Campbell 1959, 166; Luick 1914--1940, 819). Ringe and Taylor reject that analysis because *mā* 'more' and *cū* 'cow' are not plausibly unaccented words (Ringe and Taylor 2014, 86, §3.3.1). Crist formulates an Ingvaeonic rule in which *z is lost after front vowels, with compensatory lengthening, covering preconsonantal cases as well; his data contain no word-final back-vowel monosyllables, so forms like **hwaz* and the ancestor of *cū* fall outside what his statement can decide — a documented gap rather than a refutation (Crist 2002, 1, 4, §§1, 10). Kilday narrows the preconsonantal subcase to Old Saxon and Old Frisian while accepting the word-final monosyllabic loss for Old English, contrasting regular *meord* 'reward' with the loanword-influenced *mēd* 'reward' (Kilday 2024, 1–3). CAPR adopts Ringe and Taylor's quality-neutral formulation because it alone generates the back-vowel witnesses, while noting that the front-vowel forms are compatible with both analyses.

Apparent counterexamples are analogical, not phonological: OE *dēor* 'deer', *ār* 'oar', and *gār* 'spear' show final *-r* from levelling out of inflected forms where the sibilant was word-internal and regularly rhotacized, not from retention of word-final *z (Ringe and Taylor 2014, 86, §3.3.1, n. 24). The change precedes rhotacism (SC003 EAFRhotacism), which Ringe and Taylor place last in this sequence of northern developments (Ringe and Taylor 2014, 87, §3.3.1).

The corpus now witnesses this change directly. The interrogative pronoun 'who' is selected in its nominative singular masculine cell, PGmc **hwaz* (Gothic *hwas*), precisely the form Ringe and Taylor cite for this loss (Ringe and Taylor 2014, 86, §3.3.1): the rule lengthens the short nucleus and deletes the sibilant, giving **hwā*, whence OE *hwā* 'who'. The resulting back vowel never undergoes Anglo-Frisian brightening — "**hwǣ* does not exist", as Campbell puts it (Campbell 1959, sect. 125, p. 49; Brunner 1965, sect. 137 Anm. 1, p. 129) — so the derivation ends with the attested form. Other members of Ringe and Taylor's witness set remain outside the corpus because it selects oblique or plural cells for them — 'cow' and 'meed', for instance, enter the cascade in inflected forms whose *z, where present, is word-internal.

## 21.2 SC097. Northern monosyllabic final *z-loss (MonosyllabicFinalZLoss)

```
define MonosyllabicFinalZLoss [
    {*a} -> {*ā}, {*á} -> {*ā},
    {*e} -> {*ē}, {*é} -> {*ḗ},
    {*i} -> {*ī}, {*í} -> {*ī́},
    {*o} -> {*ō}, {*ó} -> {*ō},
    {*u} -> {*ū}, {*ú} -> {*ū}
        || .#. [EnglishStarConsonant | EnglishPalatalConsonant]*
            _ {*z} .#.
] .o. [
    {*z} -> 0 ||
        .#. [EnglishStarConsonant | EnglishPalatalConsonant]*
```

```
            EnglishStarVocalic+ _ .#.
];
```

The rule first lengthens a short nucleus standing immediately before word-final **z* in a monosyllable, then deletes the **z* after any vowel in a monosyllable. The principal synthetic controls run the change on Ringe and Taylor's own witnesses, fed to the rule in their chronologically correct intermediate shapes. Short-nucleus inputs show both halves of the change at once: **hwaz* yields **hwā* and **hiz* yields **hī*, each with loss of the sibilant and compensatory lengthening (Ringe and Taylor 2014, 86, §3.3.1). A form whose nucleus is already bimoric skips the lengthening step and simply loses the sibilant: **maiz* yields **mai* at this stage, with its diphthong intact; the attested OE *mā* 'more' arises only later, when the stressed monophthongization (SC004 EAFAiMonophthongization) takes **ai* to **ā*. The word 'cow', whose history Ringe and Taylor themselves print with question marks and whose analysis remains disputed, is deliberately not used as a principal control; a long-vowel input of that shape (**kūz* yielding **kū*) merely repeats what **maiz* already demonstrates, and the word's evidentiary weight is discussed in the historical dossier rather than leaned on here.

The corpus derivation of *hwā* 'who' now fixes this rule's position empirically as well as philologically. It stands after SC020 EAFFinalZDeletion, since the two losses are historically distinct changes with the unstressed loss earlier (Ringe and Taylor 2014, 44–45, §3.1.1). Rhotacism (SC003 EAFRhotacism) follows in the executable cascade as in the historical account: Ringe and Taylor place rhotacism after this loss, at the end of the sequence of *z-losses (Ringe and Taylor 2014, 87, §3.3.1), and the cascade composes it immediately after this rule, so a sibilant removed here can never surface as *-r* — were the order reversed, **hwaz* would rhotacize to **hwar* and 'who' could never be derived; the negative controls below confirm this. Consonant-final monosyllables are untouched: their nominative **-z*, where it ever existed, was eliminated before Proto-West Germanic under SC096 RootNounNomZLoss. Word-internal *z, as in PGmc **déuzą* 'deer' on its way to OE *dēor* 'deer', does not meet the environment of this rule and duly rhotacizes.

# 22 West Germanic rhotacism

## 22.1 Historical discussion

Hogg states that Germanic *z yielded *r in intervocalic position in Old English, while final *z was generally lost (Hogg 1992, 37). Ringe and Taylor argue that this merger of *z with *r was independent in Norse and West Germanic and belongs after the Proto-West-Germanic stage (Ringe and Taylor 2014, 52, 98, 102). Crist likewise places rhotacism after earlier West Germanic *z-deletion rules and rejects treating it as an inherited Proto-Northwest-Germanic innovation (Crist 2001, 104–6; 2002, 1, 4).

The internal identifier SC003 EAFRhotacism places the change in CAPR's Early Anglo-Frisian corridor, the operational post-Proto-West-Germanic stage on the English line; historically the change is a West Germanic rhotacism, later than Proto-Germanic. It is also distinct from SC020 EAFFinalZDeletion, which removes final *z before the surviving medial consonant becomes *r.

## 22.2 SC003. West Germanic rhotacism (`EAFRhotacism`)

```
define EAFRhotacism [
    {*z} -> {*r} || EnglishStarVocalic _ ?
];
```

Breaking supplies the decisive upper boundary. If rhotacism is delayed until after SC044 OEBreaking, PGmc **líznōjanq̨* 'learn' yields †*lirnian* rather than expected OE *liornian* 'learn', PGmc **líznōθi* 'learns' yields †*lirnaþ* rather than expected *liornaþ* 'learns', PGmc **líznô* 'learn' yields †*lirna* rather than expected *liorna* 'learn', and PGmc **mízdai* 'meed' yields †*merde* rather than expected OE *meorde* 'meed'. Moving rhotacism earlier within the tested range changes no output.

The lexical evidence thus supplies a terminus ante quem but no terminus post quem. The lower boundary rests on the historical analyses cited above: Ringe and Taylor put rhotacism at the end of the sequence of *z-losses — "first *z was lost in a variety of environments …, then all surviving *z became *r" (Ringe and Taylor 2014, 87, §3.3.1) — and Crist observes that the deletions distinguish *z from *r and so must precede the merger (Crist 2002, 2–3). The rule is accordingly ordered after the three final-*z losses (SC096 RootNounNomZLoss, SC020 EAFFinalZDeletion, and SC097 MonosyllabicFinalZLoss) and before breaking, so that the derivation follows the reconstructed chronology.

# 23 Unstressed **o*-raising

## 23.1 Historical discussion

In the Northwest Germanic period an unstressed **o* was raised to **u* when **u* followed in the next syllable. The clearest case is the *n*-stem accusative singular, where inherited **-onų* was regularly raised to **-unų* before the following high vowel; Campbell treats the resulting unstressed *u* and *o* alternations for Old English (Campbell 1959, 155–56, §§373–374). The change is a general development of the unstressed suffix, not tied to any single lexeme.

## 23.2 SC021. Raising of unstressed **o* before later **u* (PNWGmcUnstressedORaising)

```
define PNWGmcUnstressedORaising [
    {*o} -> {*u} || EnglishStarVocalic EnglishStarConsonant+ _
     ↪  EnglishStarConsonant* {*ų}
];
```

No word in the present selected corpus supplies this environment: no selected input carries an unstressed **o* before a following **ų*, so SC021 PNWGmcUnstressedORaising fires in no current derivation. The earlier witness *heofon* 'heaven' no longer applies here: its selected input is now Ringe and Taylor's northern West Germanic **hebun* 'heaven', which already carries the generalized labial and contains no unstressed **o* before **ų* (Ringe and Taylor 2014, 287). The rule is retained as a genuine Northwest Germanic change, but it is presently unwitnessed and boundary-limited: no lexical form now constrains its position within the Old English sequence, and moving it earlier or later leaves every output unchanged.

One candidate witness was examined and declined. The early Northumbrian accusative *galgu* 'gallows' (Ruthwell Cross) stands among the "very few" *n*-stem forms preserving a *u*-vowel in this ending, and if its *-u* continued raised **-unų* it would witness this change directly. But Ringe and Taylor themselves hedge the form — "masc. acc. galgu is not necessarily relevant", since its *-u* may instead connect with the Old High German masculine accusative singular ending *-un* ~ *-on* on Bammesberger's analogical account (Ringe and Taylor 2014, 62–63; Bammesberger 1990, 169) — they judge the raising hypothesis phonetically sensible but resting on "too small a basis" (Ringe and Taylor 2014, 63), and they conclude of the competing analyses of these *n*-stem relics that "a decisive choice between those alternatives does not seem possible" (Ringe and Taylor 2014, 164). A single dialectally marked inflected relic whose ending admits an analogical source cannot responsibly anchor a sound law, so the rule remains unwitnessed and its coverage status is recorded as a research issue.

# 24 *mn*-dissimilation

## 24.1 Historical discussion

In the inherited *n*-stem paradigm the zero-grade oblique cells brought *m* and *n* into direct contact, and in that adjacent cluster the labial nasal dissimilated to a labial spirant: *mn* > *βn* (surfacing as *fn*). Old Norse preserves the older paradigmatic distribution, with the labial confined to the oblique cluster (*himinn* 'heaven' beside dative *hifni*); Old English and Old Saxon generalized it. Fulk treats the cluster change for early Germanic (Fulk 2018, sect. 6.14, p. 121), and the relevant *heofon* 'heaven' and *mōnaþ* 'month' material is discussed by Campbell (Campbell 1959, 189, 195, §§470, 484).

The pattern is historically established, but the lexical evidence does not constrain its position.

## 24.2 SC022. Dissimilation of adjacent *mn* (`PNWGmcMnDissimilation`)

```
define PNWGmcMnDissimilation [
    {*m} -> {*β}
        || EnglishStarVocalic _ {*n}
];
```

The rule fires only where *m* stands directly before *n*. It supplies the labial of *stefn* 'stem, trunk' from the *mn*-cluster of the *stamn*-family, and it is the historical change behind the labial of *heofon* 'heaven', which was generalized from the oblique cluster into the vowel-bearing stem before the Old English vocalic changes. (An earlier cross-syllable formulation that labialized an intervocalic *m* before a later nasal has been retired: it simulated paradigm levelling rather than a sound law.)

Moving SC022 PNWGmcMnDissimilation earlier or later leaves every output unchanged. Its place among the early consonantal changes rests on the handbook account of *mn*-dissimilation.

# 25 N-stem *n*-loss

## 25.1 Historical discussion

The broader history is the reduction and leveling of older n-stem endings in West Germanic. Ringe and Taylor describe the resulting syncretism in the n-stems, which is the wider morphological setting for the narrower step isolated here (Ringe and Taylor 2014, 72).

The path to *dōn* 'do' provides the clearest witness, but the change remains narrow in scope.

## 25.2 SC023. Loss of n-stem **n* in final position (`PNWGmcNStemNLoss`)

```
define PNWGmcNStemNLoss [
    {*ō} {*n} -> {*ǭ} || _ .#.
];
```

After SC047 OEHeavySyllableNasalApocope, PGmc **dōnǭ* 'do' fails entirely (+?) instead of yielding expected OE *dōn* 'do'; earlier placement changes no output. Thus SC023 PNWGmcNStemNLoss must feed the later apocope.

This failed derivation supplies a terminus ante quem, while the lower boundary remains unattested.

# 26 Long *ē*-lowering

## 26.1 Historical discussion

The later West Saxon forms *sċēap* ‘sheep’ and *ġēar* ‘year’ imply an earlier lowering of long *ē* before the palatal diphthongal outcomes described more fully later in the sequence. Campbell and Ringe and Taylor discuss those later West Saxon outputs directly (Campbell 1959, 69–70, §185; Ringe and Taylor 2014, 215–16, §6.5.1).

The change is historically recognizable, but the lexical evidence establishes only a later boundary.

## 26.2 SC024. Lowering of long *ē* before non-nasal consonants (PNWGmcLongELowering)

```
define PNWGmcLongELowering [
    {*ē} -> {*ǣ} || _ [EnglishStarConsonant - EnglishStarNasal],
    {*ḗ} -> {*ǣ} || _ [EnglishStarConsonant - EnglishStarNasal]
];
```

After SC056 OEWsPalatalDiphthongization, long *ē* > *ǣ* can no longer produce the expected West Saxon forms: PGmc **skḗpą* ‘sheep’ yields †*sċīep* rather than OE *sċēap* ‘sheep’, and PGmc **jḗrą* ‘year’ yields †*ġīer* rather than *ġēar* ‘year’. Earlier placement changes no output, so SC024 PNWGmcLongELowering has a secure upper boundary.

Its lower boundary remains a matter of handbook chronology.

# 27 Long *ē* nasal-rounding

## 27.1 Historical discussion

Before nasals, older long *ē* can round toward the *ō*-vocalism seen later in *mōnaþ* ‘month’ and *mōna* ‘moon’ / *mōn* ‘moon’-type material. Campbell treats this split directly in his discussion of Germanic long *ē* before nasal consonants (Campbell 1959, 53, §129).

The change is historically recognizable, but the tested forms supply no close relative chronology.

## 27.2 SC025. Rounding of long *ē* before nasals (PNWGmcLongENasalRounding)

```
define PNWGmcLongENasalRounding [
    {*ē} -> {*ō} || _ EnglishStarNasal,
    {*ḗ} -> {*ō} || _ EnglishStarNasal
];
```

Reversing SC025 PNWGmcLongENasalRounding with neighboring changes leaves every output unchanged. Its position beside the other *ē*-developments therefore follows the handbooks.

# 28 Nasal spirant changes

## 28.1 Historical discussion

The two rules state successive phases of a single development. Campbell describes nasal loss before voiceless spirants with compensatory lengthening and nasalization of the preceding vowel. Ringe and Taylor assign the same outcomes to inherited northern West Germanic, before late Old English (Campbell 1959, 47, §121; Ringe and Taylor 2014, 140–41).

SC026 EAFNasalSpirantLengthening adjusts the vowel while the nasal-plus-spirant sequence remains present; SC027 EAFNasalSpirantLoss then removes the nasal. The first rule must therefore precede the second.

## 28.2 SC026. North Sea Germanic nasal-spirant lengthening (EAFNasalSpirantLengthening)

```
define EAFNasalSpirantLengthening [
    {*a} -> {*ō} || _ EnglishStarNasal EnglishStarVoicelessFricative,
    {*e} -> {*ē} || _ EnglishStarNasal EnglishStarVoicelessFricative,
    {*i} -> {*ī} || _ EnglishStarNasal EnglishStarVoicelessFricative,
    {*o} -> {*ō} || _ EnglishStarNasal EnglishStarVoicelessFricative,
    {*u} -> {*ū} || _ EnglishStarNasal EnglishStarVoicelessFricative,
    {*æ} -> {*ē} || _ EnglishStarNasal EnglishStarVoicelessFricative,
    {*á} -> {*ō} || _ EnglishStarNasal EnglishStarVoicelessFricative,
    {*é} -> {*ḗ} || _ EnglishStarNasal EnglishStarVoicelessFricative,
    {*í} -> {*ī} || _ EnglishStarNasal EnglishStarVoicelessFricative,
    {*ó} -> {*ō} || _ EnglishStarNasal EnglishStarVoicelessFricative,
    {*ú} -> {*ū} || _ EnglishStarNasal EnglishStarVoicelessFricative
];
```

All three witnesses require the vowel adjustment while the nasal is still present. If SC026 EAFNasalSpirantLengthening follows SC027 EAFNasalSpirantLoss, PGmc **fúnxstiz* 'fist' yields †*fyst* rather than expected OE *fȳst* 'fist', PGmc **gánsz* 'goose' yields †*ġeas* rather than expected *gōs* 'goose', and PGmc **júgunθ* 'youth' yields †*ġeogoþ* rather than expected *ġeoguþ* 'youth'. Earlier placement changes no output. The evidence requires lengthening to precede nasal loss without supplying a lower boundary, in agreement with the handbook treatment of the two as successive phases.

## 28.3 SC027. North Sea Germanic nasal-spirant loss (EAFNasalSpirantLoss)

```
define EAFNasalSpirantLoss [
    EnglishStarNasal -> 0 || _ EnglishStarVoicelessFricative
];
```

The converse test fixes the same boundary: placing SC027 EAFNasalSpirantLoss before SC026 EAFNasalSpirantLengthening produces the same errors in *fȳst* 'fist', *gōs* 'goose', and *ġeoguþ* 'youth'. Later placement changes no output. These forms prove that the vowel was adjusted before the nasal disappeared; they provide no upper boundary for the loss.

# 29 Preconsonantal *x-loss

## 29.1 Historical discussion

Campbell explicitly treats loss of *x* and gives forms such as *fléam* ‘flight’ and *hēla* ‘heel’ as examples of the same broad development (Campbell 1959, 186, §461).

The historical evidence is firmer than the chronology: the lexical evidence does not constrain the rule’s position.

## 29.2 SC028. Loss of preconsonantal *x (`PNWGmcPreconsonantalXLoss`)

```
define PNWGmcPreconsonantalXLoss [
    {*x} -> 0 || _ {*s} EnglishStarConsonant
];
```

No witness word dates preconsonantal **x*-loss before **s* plus another consonant: moving SC028 PNWGmcPreconsonantalXLoss in either direction leaves every output unchanged. Its position within this stretch therefore rests on the handbook chronology for *x*-loss.

# 30 Anglo-Frisian ai-monophthongization

## 30.1 Historical discussion

Inherited stressed **ái* monophthongized to **ā* across the North Sea Germanic area. Ringe and Taylor place the monophthongization of **ai* among the widespread early vowel developments of the English line (Ringe and Taylor 2014, 40–41). Versloot shows that the stressed development spread in successive waves through a dialect continuum, with Old English among the widest to carry it (Versloot 2017, 281–324). The Old English **ā* is later fronted to *ǣ* in the relevant environments.

The change is areal in character. It is shared with Frisian, and its spread through the continuum gives it a range of dates and no single sharp moment.

All twenty-four corpus witnesses carry stressed **ái*; loam **láimą* 'loam' is one of them, stressed in its Old English protoform. The unstressed development **ai* > **ē* is the separate earlier change SC014 PNWGmcUnstressedAiMonophthongization.

## 30.2 SC004. Anglo-Frisian ai-monophthongization (EAFAiMonophthongization)

```
define EAFAiMonophthongization [
    {*ái} -> {*ā}
];
```

The soul form fixes the relation to interstress raising. If the monophthongization is delayed until after that change, PGmc **sáiwalō* 'soul' yields †*sāwel* rather than expected OE *sāwol* 'soul'. An earlier placement changes no output. This shows that SC004 EAFAiMonophthongization must come before SC036 OEInterStressRaising in the modeled sequence.

The unstressed development **ai* > **ē* in final and nonfinal syllables is a separate and earlier change; see SC014 PNWGmcUnstressedAiMonophthongization.

# 31 Chapter 3. From Anglo-Frisian to Old English

## 31.1 Historical interval

This chapter covers the sound changes that occurred within the Old English period: the changes that produced attested Old English from the prehistoric English forms that emerged from the Anglo-Frisian stage. The starting point is the end of the Anglo-Frisian changes of Chapter 2; the ending point is attested West Saxon Old English, the primary dialect of the CAPR corpus.

## 31.2 Scope and dialect variation

Not every change in this chapter has pan-Old-English scope. Some changes — most notably West Saxon palatal umlaut (SC060), the back-mutation rules (SC059), and the West Saxon diphthong chain (SC031–SC034) — are specifically West Saxon or more broadly southern Old English phenomena. (The West Saxon palatal-glide effects, SC016, execute early in the cascade and are treated in Chapter 1.)

The CAPR derivations target West Saxon Old English citation forms as the default comparator. Changes that belong to other dialects, or that are absent from West Saxon, may appear in lexical entries as comparanda rather than as derivational steps.

The existing reader-facing sound-change sections record which changes have pan-Old-English scope and which are specifically West Saxon or Anglian (Campbell 1959, sects 1–10; Hogg 1992, sects 1.1–1.15).

## 31.3 Chapter structure

The changes in this chapter fall into several natural historical subgroups, though the boundaries between them are not always sharp:

Early Old English changes linked to the Anglo-Frisian inheritance: Changes that feed directly on, or are closely related to, Anglo-Frisian brightening (SC043), whose section opens the vowel corridor of this chapter. The `*awj` glide formation (SC029), `*au` fronting (SC030), and the West Saxon diphthong chain (SC031–SC034) all operate on the vowel inventory shaped by Anglo-Frisian brightening. OE Breaking (SC044) and a-Restoration (SC046) similarly presuppose the fronted *æ input.

Old English consonantal changes: Velar palatalization (SC052), palatalization of `*sk` (SC051), j-cluster coalescence (SC057), and related changes produce the characteristically Old English consonant phonemes. Hogg discusses these as OE consonant changes that are not broadly West Germanic (Hogg 1992, sects 7.18–7.23).

Old English i-umlaut and its context: The i-umlaut (SC055) is one of the most productive changes in the Old English nominal and verbal morphology. Its relative chronology in relation to breaking, palatalization, and back-mutation is carefully documented in the existing CAPR chronology evidence audit and individual dossiers (Campbell 1959, sects 193–204; Hogg 1992, sects 5.62–5.68).

Late Old English syllabic reduction and apocope: High-vowel apocope (SC063), medial syncope (SC065), and the cluster of late unstressed-vowel changes (SC069–SC078) represent the later stage of Old English phonological history, when the syllabic structure of the language began to shift toward the more reduced profile of Middle English.

## 31.4 Cascade positions and historical order

Chapters in this part of the book follow the executable cascade order, which models the reconstructed chronology, so every section in this chapter appears at its cascade position. A few rules presented here carry stage labels from earlier periods; their individual sections discuss the label and its history, and a later renaming pass will resolve the residue:

- SC041 (PWGmc Final Bare-`*a` Loss) and SC042 (Surviving Bimoric `*ō` Unrounding) carry Proto-West Germanic labels but execute in this stretch of the cascade.
- SC064 (NWGmc `*-n` Stem `*n` Loss) carries a Northwest Germanic label but executes after OE High-Vowel Apocope (SC063).
- SC049 (PGmc B Allophony) carries a Proto-Germanic label but executes here.

## 31.5 Sources

Campbell's *Old English Grammar* is the primary source for the dating and scope of individual changes in this chapter (Campbell 1959). Hogg's *Grammar of Old English* provides modern reassessments and additional relative-chronology evidence (Hogg 1992). Ringe and Taylor supply the most detailed relative-chronology analysis for the earlier portion of the chapter, through back-mutation (Ringe and Taylor 2014, 70–160). Fulk's *Comparative Grammar* provides additional coverage for morphological conditioning (Fulk 2018). For individual changes, source-specific citations appear in the relevant sound-change sections.

# 32 Awj glide formation and au-fronting

## 32.1 Historical discussion

The *hīeġ* 'hay' and *strīeġan* 'strew' material undergoes both changes. Glide formation reshapes the older *awj* sequence, and fronting then affects the resulting *au*. Campbell's discussion of these outcomes and Ringe and Taylor's derivations of *hīeġ* and *strīeġan* describe the same sequence (Campbell 1959, 46, §120; Ringe and Taylor 2014, 188).

Glide formation creates the input to fronting; diphthong leveling follows both.

## 32.2 Historical discussion of awj glide formation

Older *awj* sequences are the source of forms such as *hīeġ* 'hay' and *strīeġan* 'strew'. Campbell treats the relevant developments directly, and Ringe and Taylor likewise trace the same material through intermediate *auj*-type stages (Campbell 1959, 46, §120; Ringe and Taylor 2014, 188).

The sources establish glide formation, while the witness forms supply only a later boundary.

## 32.3 SC029. Glide formation in **awj* (`OEAwjGlideFormation`)

```
define OEAwjGlideFormation [
    {*á} {*w} {*w} {*j} -> {*áu} {*j},
    {*a} {*w} {*w} {*j} -> {*au} {*j},
    {*á} {*w}      {*j} -> {*áu} {*j},
    {*a} {*w}      {*j} -> {*au} {*j}
];
```

The *hīeġ* 'hay' and *strīeġan* 'strew' derivations show that *awj* reshaping prepared the input to fronting. If fronting is applied first, PGmc **xáwwją* 'hay' yields †*hauġ* rather than expected OE *hīeġ* 'hay', and PGmc **stráwjaną* 'strew' yields †*strauian* rather than expected *strīeġan* 'strew'. Earlier placement of glide formation changes no output, so these forms supply an upper boundary without a corresponding lower one.

## 32.4 Historical discussion of au-fronting

Once the glide sequence is in place, *au*-fronting produces the fronted diphthongal outcomes of the broader West Saxon vowel history. Campbell describes *au* > *ēa* (Campbell 1959, 53–54, §135).

Fronting must follow glide formation and precede diphthong leveling, which applies to a wider set of derivations.

## 32.5 SC030. Fronting of **au* (`OEAuFronting`)

```
define OEAuFronting [
    {*au} -> {*aeu},
    {*áu} -> {*áeu}
];
```

Two distinct failure sets confine fronting. Placed before glide formation, it produces the wrong forms: PGmc **xáwwją* 'hay' yields †*hauġ* rather than expected OE *hīeġ* 'hay', and PGmc **stráwjaną* 'strew' yields †*strauian* rather than expected *strīeġan* 'strew'. Placed after diphthong leveling, PGmc **galáubijaną* 'believe', **bráudą* 'bread', and **dráugmaz* 'dream', together with sixteen other derivations, fail to produce output at all (+*?*) instead of yielding expected OE *ġelīefan* 'believe', *brēad* 'bread', and *drēam* 'dream'. The lexical errors require fronting to follow glide formation, while the failed derivations require it to precede diphthong leveling.

The later failure set consists of failed derivations, not competing Old English surface forms.

# 33 West Saxon diphthong sequence

## 33.1 Historical discussion

Four distinct developments shape the West Saxon diphthongal field. Campbell discusses inherited *aw*/*ew* outcomes, palatal-triggered diphthongization, and later Anglian smoothing in connected but separate parts of the vowel history; Hogg likewise distinguishes the palatal-diphthongal developments (Campbell 1959, 46, 53–54, 65–70, 95–96, §§120, 135–136, 170–176, 185, 223–227; Hogg 1992, 106–7, 111–12).

The closest interaction joins *ww*-simplification and long-*aw* diphthongization, which together shape *dēaw* 'dew' and *hēawan* 'hew'. Diphthong leveling regularizes a wider field, while long-*ew* diphthongization carries *ēow* into the later environment of breaking.

## 33.2 Historical discussion of WW simplification

West Germanic *ww* sequences lie behind forms such as *dēaw* 'dew' and *hēawan* 'hew', and Campbell treats them as part of the early West Germanic diphthong history (Campbell 1959, 46, §120).

SC031 OEWWSimplification precedes the later long-diphthong outcomes.

## 33.3 SC031. Simplification of **ww* sequences (`OEWWSimplification`)

```
define OEWWSimplification [
    {*w} {*w} -> {*w}
];
```

The *dēaw* 'dew' and *hēawan* 'hew' derivations establish that doubled *w* was simplified before the long *ēaw* development. If SC031 OEWWSimplification follows SC034 OEAwLongDiphthong, PGmc **dáwwō* 'dew' yields †*dawu* rather than expected OE *dēaw* 'dew', and PGmc **xáwwaną* 'hew' yields †*hawan* rather than expected *hēawan* 'hew'. Earlier placement changes no output. The witnesses require simplification before the long-diphthong change and leave the lower boundary to the broader West Saxon chronology.

## 33.4 Historical discussion of diphthong leveling

Forms such as *hēafod* 'head' reflect the redistribution of diphthongal outcomes across a wider set of words. Campbell describes smoothing and related later monophthongization, although the rule below is more narrowly conditioned than any single textbook label (Campbell 1959, 95–96, §§223–227).

The evidence for SC032 OEDiphthongLeveling is less self-contained than that for the *dēaw* 'dew' / *hēawan* 'hew' developments.

## 33.5 SC032. Leveling of diphthongal outputs (`OEDiphthongLeveling`)

```
define OEDiphthongLeveling [
    {*aeu} -> {*ēa},
    {*áeu} -> {*ēa},
    {*eu} -> {*ēo},
    {*éu} -> {*ēo},
    {*iu} -> {*ēo},
    {*íu} -> {*ēo},
    {*e} {*u} -> {*eo},
    {*é} {*u} -> {*éo},
    {*i} {*u} -> {*eo}
];
```

The two edges of this interval fail differently. Before SC030 OEAuFronting, PGmc **galáubijaną* 'believe', **báug* 'bow', and **bráudą* 'bread' produce no output (+?) instead of expected OE *ġelīefan* 'believe', *bēag* 'bow', and *brēad* 'bread', alongside fifteen other failed derivations. After SC040 OEMedUnstressedULowering, PGmc **xáubudą* 'head' yields †*hēafud* rather than expected *hēafod* 'head'. Absence at the lower edge places diphthong leveling after fronting; the wrong surface form at the upper edge places it before medial unstressed-*u* lowering.

## 33.6 Historical discussion of long *ēow*

The long *ēow* forms of *ċēowan* 'chew', *fēower* 'four', and *cnēow* 'knee' form part of the West Saxon vowel history, although their clearest ordering relation points forward. Campbell describes early *eu* in Old English, and Ringe and Taylor give the corresponding examples from chew, four, and knee (Campbell 1959, 53–54, §136; Ringe and Taylor 2014, 188, 202).

The only boundary established by the lexical evidence for SC033 OEEwLongDiphthong lies ahead at SC044 OEBreaking.

## 33.7 SC033. Long *ēow* before following vowels and weak endings (OEEwLongDiphthong)

```
define OEEwLongDiphthong [
    {*e} {*w} -> {*ēo} {*w} || _ OEEwLongContext,
    {*i} {*w} -> {*ēo} {*w} || _ OEEwLongContext,
    {*é} {*w} -> {*ēo} {*w} || _ OEEwLongContext,
    {*í} {*w} -> {*ēo} {*w} || _ OEEwLongContext
];
```

The long *ēow* of *ċēowan* 'chew', *fēower* 'four', and *cnēow* 'knee' supplies only a terminus ante quem. If SC033 OEEwLongDiphthong follows SC044 OEBreaking, PGmc **kéwwaną* 'chew' yields †*ċeowan* rather than expected OE *ċēowan* 'chew', PGmc **fédwōr* 'four' yields †*feower* rather than expected *fēower* 'four', and PGmc **knéwą* 'knee' yields †*cneow* rather than expected *cnēow* 'knee'. Earlier placement changes no output. The sources associate *ew* and *iw* with the same diphthongal history but furnish no lower boundary.

## 33.8 Historical discussion of long *ēaw*

After SC031 OEWWSimplification has reduced *ww* to single *w*, the remaining *aw* sequence can develop into the long *ēaw* seen in *dēaw* 'dew' and *hēawan* 'hew'. Campbell treats these outputs in the early diphthong history of West Germanic and Old English (Campbell 1959, 46, 53–54, §§120, 135–136). The resulting long diphthong is *ēaw*.

SC034 OEAwLongDiphthong follows SC031 OEWWSimplification locally and must also precede SC043 EAFBrightening.

## 33.9 SC034. Long *ēaw* before following vowels (`OEAwLongDiphthong`)

```
define OEAwLongDiphthong [
    {*a} {*w} -> {*ēa} {*w} || _ [EnglishStarVocalic | {*ô}],
    {*á} {*w} -> {*ḗa} {*w} || _ [EnglishStarVocalic | {*ô}]
];
```

A local feeding relation and a later vowel change confine *aw* > *ēaw*. Before SC031 OEWWSimplification, PGmc **dáwwō* 'dew' yields †*dawu* rather than expected OE *dēaw* 'dew', and PGmc **xáwwaną* 'hew' yields †*hawan* rather than expected *hēawan* 'hew'. After SC043 EAFBrightening, PGmc **skáwōjaną* 'show' yields †*sċawian* rather than expected OE *sċēawian* 'show', PGmc **skáwōθi* 'shows' yields †*sċawaþ* rather than expected *sċēawaþ* 'shows', and PGmc **stráwą* 'straw' yields †*stræw* rather than expected *strēaw* ‘straw’. The *dēaw* and *hēawan* forms require long-diphthong formation after simplification, while *sċēawian* requires it before brightening; the handbooks assign the same interval to the West Saxon development.

# 34 Prefix and compound adjustments

## 34.1 Historical discussion of prefixal *a-reduction

Weakly stressed prefixes can lose their older low vowel early in Old English, and that is the historical setting for SC035 OEPrefixAReduction. Campbell treats the small class of pretonic losses directly, while Ringe and Taylor's derivation of **galaubijana* 'believe' supplies the comparative witness for the same development (Campbell 1959, 147, §354; Ringe and Taylor 2014, 245; 2014, 267).

The rule has a narrow historical range and gives prefixed forms the weak vowel inherited by later vocalic changes.

## 34.2 SC035. Reduction of prefixal *a (`OEPrefixAReduction`)

```
define OEPrefixAReduction [
    {*a} -> {*ĕ}
        || .#. {*g} _
           [EnglishStarConsonant | EnglishPalatalConsonant]
           EnglishStarVocalic
];
```

The prefix of *ġelīefan* 'believe' supplies the upper boundary for **ga-* > **ge-*. If SC035 OEPrefix-AReduction follows SC043 EAFBrightening, PGmc **galáubijaną* 'believe' yields †*ġealīefan* rather than expected OE *ġelīefan* 'believe'. Earlier placement changes no output, so the witness dates prefix reduction before brightening without locating its beginning.

## 34.3 Historical discussion of inter-stress raising

SC036 OEInterStressRaising has the strongest evidence of the three. Campbell's discussion of *weorold* 'world' / *weoruld* 'world' and Ringe and Taylor's derivation of **weraldu* 'world' > **weruldu* 'world' > OE *weorold* place the rule squarely in the history of low-stress medial vowels (Campbell 1959, 141–42, §§338–339; Ringe and Taylor 2014, 322, §6.3.3).

The rule changes the vowel between stronger stress peaks, and its witnesses consequently constrain the relative chronology.

## 34.4 SC036. Raising of medial *a between stress peaks (OEInterStressRaising)

```
define OEInterStressRaising [
    {*a} -> {*u}
        || PGmcStarVowel EnglishStarConsonant* _
           [EnglishStarConsonant - {*j}]+ [{*u}|{*ū}],
    {*à} -> {*u}
];
```

The two boundaries have unequal force. Before SC019 PNWGmcFinalLongORaising, PGmc **sáiwalō* 'soul' yields †*sāwel* rather than expected OE *sāwol* 'soul'; after SC040 OEMedUnstressedU-Lowering, it yields †*sāwul* rather than *sāwol*, while PGmc **wír-àldu* 'world' yields †*weoruld* rather than *weorold* 'world'. The distant lower boundary places inter-stress raising after final long-*o* raising, and the local upper boundary places it before medial unstressed-*u* lowering. In handbook terms, medial **a* > **u* belongs to the *world-* and *soul-*type low-stress vocalism that followed the earlier final-vowel changes.

## 34.5 Historical discussion of compound linking syncope

Compound members with weakened force often lose or reshape their linking vowels, and Campbell treats that broad pattern through reduced second elements, connecting vowels, and obscured compounds (Campbell 1959, 148–49, §§356–357; Campbell 1959, 153, §367; Campbell 1959, 159, §§386–387).

SC037 OECompoundLinkingSyncope captures this pattern in compounds such as *reġnboga* 'rainbow'. The only boundary the lexical evidence supplies is the immediately following technical stress-stripping stage, which is not a sound change.

## 34.6 SC037. Syncope of compound linking vowels (OECompoundLinkingSyncope)

```
define OECompoundLinkingSyncope [
    [{*a}|{*i}|{*u}] -> 0
        || PGmcStarAcuteVowel OEAnyConsonant+ _
           OEAnyConsonant+ PGmcStarGraveVowel
];
```

The *reġnboga* 'rainbow' test exposes a bookkeeping dependency rather than a historical sound-change boundary. After SC038 OEStripSecondaryStress, PGmc ***régna-bùgô*** 'rainbow' yields †*reġnefoga* rather than expected OE *reġnboga* 'rainbow', because the technical stage has erased the stress information that licenses syncope. The handbooks instead place weakened compound junctures with the behavior described under SC035 OEPrefixAReduction and SC036 OEInterStressRaising.

# 35 Medial unstressed vowel changes

## 35.1 Historical discussion

The history of *wuduwe* ‘widow’ orders these two changes within the same low-stress vocalic development. Campbell discusses both the *w*-conditioned *u* forms and the later *weorold* ‘world’ / *weoruld* ‘world’ alternation, while Ringe and Taylor give the same connection comparatively in **widuwon-*, **weraldu* ‘world’, and **jugunþi* ‘youth’ (Campbell 1959, 92, §218; Campbell 1959, 140, §332; Campbell 1959, 141–42, §§338–339; Ringe and Taylor 2014, 267; Ringe and Taylor 2014, 322, §6.3.3).

SC039 OEWICombinativeUUmlaut feeds the vowel sequence subsequently reshaped by SC040 OEMedUnstressedULowering. Initial *w* conditions the first change.

## 35.2 SC039. Combinative **u*-umlaut in *wi*-forms (`OEWICombinativeUUmlaut`)

```
define OEWICombinativeUUmlaut [
    {*í} -> {*ú}
        || .#. {*w} _ EnglishStarConsonant [{*u} | {*o}]
];
```

The *wuduwe* ‘widow’ derivation answers one narrow question about *wi*-forms. If SC039 OEWICombinativeUUmlaut follows SC040 OEMedUnstressedULowering, PGmc **wíduwōn* ‘widow’ yields †*wudowe* rather than expected OE *wuduwe*; earlier placement changes no output. The witness requires combinative u-umlaut to precede medial lowering and supplies no lower boundary.

## 35.3 SC040. Lowering of medial unstressed **u* (OEMedUnstressedULowering)

```
define OEMedUnstressedULowering [
    {*u} -> {*o}
        || [EnglishStarVocalic - [{*u}|{*ū}|{*ú}]]
           [EnglishStarConsonant | EnglishPalatalConsonant]+ _
           [[EnglishStarConsonant | EnglishPalatalConsonant] - {*m}]
];
```

The two witnesses date medial unstressed **u* > **o* at very different scales. Before SC039 OEWICombinativeUUmlaut, PGmc **wíduwōn* ‘widow’ yields †*wudowe* rather than expected OE *wuduwe* ‘widow’; after SC072 OEUnstressedLongVowelShortening, PGmc **júgunθ* ‘youth’ yields †*ġeogoþ* rather than expected *ġeoguþ* ‘youth’. The local *weorold* ‘world’ and widow evidence places lowering after combinative u-umlaut, while the youth form supplies only the distant requirement that lowering precede unstressed long-vowel shortening.

# 36 Final bare-*a* loss

## 36.1 Historical discussion

I isolate the loss of final short low vowels within the broader erosion of final syllables described by the handbooks (Campbell 1959, 143, §341; Ringe and Taylor 2014, 60–61).

Final bare-a loss follows the medial unstressed vowel changes and precedes restoration, which depends on the environment left by the loss.

## 36.2 SC041. Loss of final bare **a* (`PWGmcFinalBareALoss`)

```
define PWGmcFinalBareALoss [
    {*a} -> 0 || _ .#.
];
```

The two sides of final bare-*a* loss rest on different evidence. Applied before final *z*-deletion, the change gives the wrong outputs: PGmc **bárdaz* ‘beard’ yields †*bearda* rather than expected OE *beard* ‘beard’, and PGmc **kámbaz* ‘comb’ yields †*camba* rather than expected *camb* ‘comb’. Applied after restoration, PGmc **kráftaz* ‘craft’ yields †*craft* rather than expected OE *cræft* ‘craft’, and PGmc **dágaz* ‘day’ yields †*dag* rather than expected *dæġ* ‘day’. The distant lower limit follows final *z*-loss; the local feeding relation precedes restoration, which requires the environment created by the vowel loss.

# 37 Surviving bimoric **ō* unrounding

## 37.1 Historical discussion

The handbooks do not isolate a large independent sound change under this label. The surviving bimoric **ō* in the pathway to *ræste* ‘rest’ nevertheless undergoes unrounding before SC043 EAFBrightening. Campbell, Hogg, and Ringe and Taylor describe the surrounding fronting and restoration history without naming this feeder separately (Campbell 1959, 52, 60, §§131, 157–158; Hogg 1992, 101, 119; Ringe and Taylor 2014, 157–58, 189–90).

The sole witness establishes a local relation to brightening but supports no broader generalization.

## 37.2 SC042. Unrounding of the surviving bimoric **ō* (PWGmcSurvivingBimoricOUnrounding)

```
define PWGmcSurvivingBimoricOUnrounding [
    {*ō} -> {*ā} || EnglishStarVocalic [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant]+ _ .#.
];
```

The single *ræste* ‘rest’ derivation carries the chronology of bimoric **ō* > **ā*. Before SC020 EAFFinalZDeletion or after SC043 EAFBrightening, PGmc **rástōz* ‘rest’ yields †*rasta* rather than expected OE *ræste*. Unrounding must therefore follow final *z*-loss and precede brightening, although only the relation to brightening is local.

# 38 Anglo-Frisian brightening

## 38.1 Historical discussion

Anglo-Frisian Brightening or First Fronting turns low **a* into fronted **æ*-type outcomes outside nasal environments. Later Old English developments presuppose this fronted stage even where they partly conceal it. Campbell gives the classical statement of the change, Hogg supplies the standard modern labels, and Ringe and Taylor establish its local chronology with breaking and restoration (Campbell 1959, 52, §131; Hogg 1992, 101, 119; Ringe and Taylor 2014, 157–58, 189–90; Fulk 2018, 73–74, §§4.12–4.13).

Brightening creates the input to SC044 OEBreaking, while SC046 OEARestoration later partly reverses its outcome before back vowels.

## 38.2 SC043. Fronting of low **a* outside nasal environments (EAFBrightening)

```
define EAFBrightening [
    AngloFrisianBrighteningUnstressed .o.
    AngloFrisianBrighteningStressed .o.
    AngloFrisianBrighteningLongFinal
];
```

Two derivations place low **a* > **æ* between unrounding and breaking. Before SC042 PWGmc-SurvivingBimoricOUnrounding, PGmc **rástōz* 'rest' yields †*rasta* rather than expected OE *ræste* 'rest'. After SC044 OEBreaking, PGmc **sláxaną* 'slay' yields *sleaan* | *slēaan* rather than expected OE *slēan* 'slay'. The first witness requires brightening to receive the outcome of the surviving-bimoric **ō* development; the second requires breaking to receive the fronted vowel.

# 39 Breaking and velar-fricative palatalization

## 39.1 Historical discussion

Breaking creates *eo*-type outputs before *h*, *rC*, and *lC*; velar-fricative palatalization then operates in that reshaped environment. Campbell, Ringe and Taylor, and Fulk place breaking after brightening. The following fricative palatalization is more narrowly conditioned (Campbell 1959, 54, 166, §§139, 405–406; Ringe and Taylor 2014, 168–69, 213–14, §§6.2.1–6.2.3, 6.4.1–6.4.2; Fulk 2018, 73–74, §4.13).

Breaking has the fuller handbook treatment, while velar-fricative palatalization follows it locally in the *feoh* ‘cattle’ and *feohtan* ‘fight’ type derivations.

## 39.2 SC044. Breaking before *h*, *rC*, and *lC* (`OEBreaking`)

```
define OEBreaking OEBreakingA
    .o. OEBreakingE
    .o. OEBreakingI;
```

Breaking must encounter the vowel created by brightening and must precede the fricative change seen in *feoh* ‘fee’ and *feohtan* ‘fight’. Before SC043 EAFBrightening, PGmc **sláxaną* ‘slay’ yields *sleaan* | *slēaan* rather than expected OE *slēan* ‘slay’. After SC045 OEVelarFricativePalatalization, PGmc **féxu* ‘cattle’ yields †*fehu* rather than expected OE *feoh*, and PGmc **féxtaną* ‘fight’ yields †*fehtan* rather than expected *feohtan*. The two feeding relations place breaking between brightening and velar-fricative palatalization.

## 39.3 SC045. Palatalization of velar fricatives beside front vowels (OEVelarFricativePalatalization)

```
define OEVelarFricativePalatalization [
    {*x} -> {*ç} || _ EnglishStarFrontVowel,
    {*ɣ} -> {*j} || _ EnglishStarFrontVowel,
    {*x} -> {*ç} || EnglishStarFrontVowel _,
    {*ɣ} -> {*j} || EnglishStarFrontVowel _,
    {*x} -> {*ç} || _ {*j},
    {*ɣ} -> {*j} || _ {*j}
]
    .o. EnglishStarAlphabet*;
```

The local chronology comes from *feoh* ‘cattle’ and *feohtan* ‘fight’. Before SC044 OEBreaking, palatalization of **x* and **ɣ* beside front vowels or **j* makes PGmc **féxu* ‘cattle’ yield †*fehu* rather than expected OE *feoh*, and PGmc **féxtaną* ‘fight’ yield †*fehtan* rather than expected *feohtan*. The distant upper limit comes from *six* ‘six’: after SC060 OEWsPalatalUmlaut, PGmc **séxs* ‘six’ yields †*sihs* rather than expected OE *six*. Breaking therefore feeds velar-fricative palatalization directly, while palatal umlaut supplies only the broader upper limit.

# 40 A-restoration and nasal changes

## 40.1 Historical discussion of A-restoration

Campbell's restoration of *a* before following back vowels and Ringe and Taylor's later retraction describe the same post-brightening development (Campbell 1959, 60–61, §§157–159; Ringe and Taylor 2014, 189–90, §6.3.1; Fulk 2018, 74, §4.13). Some outcomes of Anglo-Frisian fronting survive only in environments where restoration does not return them to back *a*.

SC046 OEARestoration has firmer handbook support than the two following nasal rules.

## 40.2 SC046. Restoration of **a* before following back vowels (OEARestoration)

```
define OEARestoration (
    {*æ} -> {*a} || _
        OEARestorationIntervening OEARestorationTriggerVowel
        - OEARestorationIntervening OEARestorationWeakTailVowel
);
```

Restoration must receive fronted **æ* and return **a* before the nasal-tail changes. Before SC043 EAFBrightening, PGmc **bákaną* 'bake' yields †*bæcan* rather than expected OE *bacan* 'bake', and PGmc **fáraną* 'fare' yields †*færan* rather than expected *faran* 'fare'. After SC048 OESecondaryNasalization, **bákaną* again yields †*bæcan* instead of *bacan*, while PGmc **wádaną* 'wade' yields †*wædan* instead of *wadan* 'wade'. These independent witness pairs place restoration after brightening and before secondary nasalization.

## 40.3 Historical discussion of heavy-syllable nasal loss and secondary nasalization

Heavy-syllable nasal apocope removes the final nasalized vowel; secondary nasalization then marks the preceding *a* before final *n*. The handbooks do not isolate both developments under equally prominent labels. Campbell describes later nasal loss and the back-mutation environment; Ringe and Taylor provide the later relation to back mutation (Campbell 1959, 86, 166, §§205–206, 403; Ringe and Taylor 2014, 319, §6.9.4).

The reciprocal failure set fixes the order: apocope removes the ending before secondary nasalization acts on the remaining structure. Restoration receives the fuller historical treatment in the handbooks.

## 40.4 SC047. Heavy-syllable nasal apocope of final **ą* (OEHeavySyllableNasalApocope)

```
define OEHeavySyllableNasalApocope [
    {*ą} -> 0 || OEAnyConsonant _ .#.
];
```

The evidence for final nasalized **ą* loss is sharply asymmetric. Before SC034 OEAwLongDiphthong, the single PGmc witness **stráwą* 'straw' yields †*stræw* rather than expected OE *strēaw* 'straw'. After SC048 OESecondaryNasalization, PGmc **bákaną* 'bake' yields †*bacen* rather than expected OE *bacan* 'bake', and PGmc **bíndaną* 'bind' yields †*binden* rather than expected *bindan* 'bind', alongside a broad *-en* failure set. One lower witness places apocope after long-diphthong formation; many reciprocal upper failures place it before secondary nasalization.

## 40.5 SC048. Secondary nasalization before final **n* (OESecondaryNasalization)

```
define OESecondaryNasalization [
    {*a} -> {*ą} || _ {*n} .#.
];
```

The broad *-an*/*-en* split fixes the lower boundary of final **a* nasalization before *n*. Before SC047 OEHeavySyllableNasalApocope, PGmc **bákaną* 'bake' yields †*bacen* rather than expected OE *bacan* 'bake', and PGmc **bíndaną* 'bind' yields †*binden* rather than expected *bindan* 'bind'. The upper boundary comes from back mutation. After SC059 OEBackMutation, PGmc **stélaną* 'steal' yields †*steolan* rather than expected OE *stelan* 'steal', and PGmc **wébaną* 'weave' yields †*weofan* rather than expected *wefan* 'weave'. Reciprocal nasal-tail failures place secondary nasalization after apocope, and the later mutation witnesses place it before back mutation; SC046 OEARestoration retains the clearest independent historical support.

# 41 B allophony

## 41.1 Historical discussion

The positional alternation of Germanic **b* is a Proto-Germanic distributional feature. Hogg states the Old English distribution clearly: /b/ is a stop initially, after nasals, and in gemination, while the same segment is otherwise realized as a voiced bilabial fricative (Hogg 1992, 101–2). Ringe and Taylor support the broader West Germanic background by treating Proto-West-Germanic **b* as a segment whose stop and fricative values depend on position (Ringe and Taylor 2014, 121), and Luick's spelling evidence shows the same labial fricative pattern in Old English (Luick 1914--1940, 107).

The distribution is narrow, but later changes presuppose the stop-fricative alternation. CAPR implements the rule at a late cascade position for computational reasons: the alternation must interact with consonant environments shaped by intermediate rule applications. Its historical stage is Proto-Germanic.

## 41.2 SC049. Distribution of **b* after vowels and liquids (PGmcBAllophony)

```
define PGmcBAllophony [
    {*b} -> {*β} || PGmcStarVocalic _,
    {*b} -> {*β} || [{*l} | {*r}] _
] .o. [
    {*β} -> {*b} || _ {*b}
];
```

The handbooks describe **b*/**bb* as a positional alternation within the consonant system, and one compound supplies its chronological consequence. Before SC037 OECompoundLinkingSyncope, *reġnboga* 'rainbow' develops as †*reġnfoga* rather than expected OE *reġnboga*; later placement creates no comparable failure. The witness places b-allophony after compound-linking syncope without turning the alternation into an independent sound law.

# 42 Sievers-law syncope

## 42.1 Historical discussion

Sievers' Law concerns a prosodic and morphological adjustment in heavy stems. It is a distributional rule distinct from b-allophony (SC049 PGmcBAllophony). Adamczyk treats the Old English reflexes of the law as historical evidence from weak verbs and related formations (Adamczyk 2001, 61–72). Fulk gives the compact comparative summary through familiar forms such as *biddan* 'ask', *sellan* 'give', and *nerian* 'save' (Fulk 2018, 127, §6.15).

Sievers-law syncope is narrow in scope, but its relation to the following palatalization is lexically secure. Its earlier limit is less sharply defined than that of the preceding allophony rule.

## 42.2 SC050. Sievers-law syncope (`SieversLawSyncope`)

```
define SieversLawSyncope [
    {*i} -> 0 || [EnglishStarConsonant | EnglishPalatalConsonant] _ {*j}
];
```

The Sievers-law reduction **-CijV-** > **-CjV-**, including loss of **i* before **j*, must precede palatalization. If SC050 SieversLawSyncope follows SC052 OEVelarPalatalization, PGmc **strákkijaną* 'stretch' yields †*streccan* rather than expected OE *streċċan* 'stretch'; earlier placement creates no comparably precise error. The single cluster witness therefore places syncope before velar palatalization.

# 43 Palatalization of **sk* to **sc*

## 43.1 Historical discussion

The palatalization of **sk* to Old English **sc* is one of the recognizable early cluster changes in the larger palatalization zone. Campbell distinguishes the cluster from plain velars when he remarks that **sk* is especially prone to palatalization and assibilation (Campbell 1959, 278, §440). Hogg gives the same change a clearer structural place by treating **sk* beside the palatalization of plain velars and before the later West Saxon diphthongal developments (Hogg 1992, 106–7, 111–12). Ringe and Taylor make the same sequence explicit when they distinguish the earlier palatalization of velars and **sk* from the later diphthongization after already palatal consonants (Ringe and Taylor 2014, 213–16, §§6.4.1, 6.5.1).

Luick places the cluster change within a broader early movement toward palatal articulation, while still allowing later vowel consequences to form a different chapter of the history (Luick 1914--1940, 157, §168). Fulk's summary is the most concise warning against overextension: Old English **sc* is palatal except in the well-known back-vowel environments that preserve harder outcomes (Fulk 2018, 28). The result is a historically clear rule, but not an identity between the cluster change and the later umlautal developments.

## 43.2 SC051. Palatalization of **sk* to **sc* (`OESkPalatalization`)

```
define OESkPalatalization [
    {*s} {*k} -> {*ʃ} || .#. _
] .o. [
    {*s} {*k} -> {*ʃ} || EnglishStarFrontVowel _ (EnglishStarConsonant | .#.)
] .o. [
    {*s} {*k} -> {*ʃ} || (EnglishStarConsonant | .#.) _ EnglishStarFrontVowel
] .o. [
    {*s} {*k} -> {*ʃ} || _ {*j}
] .o. [
    {*s} {*k} -> {*ʃ} || {*j} _
];
```

The non-fronted vowels of *flasce* 'flask' and *wascan* 'wash' fix the lower boundary of **sk* > **sc*. Before SC046 OEARestoration, the forms are fronted too soon, yielding *flæsce* 'flask' and *wæscan* 'wash' rather than expected OE *flasce* and *wascan*. This places SC051 OESkPalatalization after restoration.

Five witnesses establish the upper boundary collectively. The palatal cluster must already underlie *sċeaft* 'shaft', *sċēar* 'shear', *sċēaþ* 'sheath', *sċēap* 'sheep', and *sċield* 'shield' before SC056 OEWsPalatalDiphthongization. The **sċea-* 'sea'/*sċie-** set therefore places cluster palatalization before the West Saxon vowel change. The cluster change occupies the same palatalization zone as SC052 OEVelarPalatalization while remaining distinct from plain-velar palatalization and the later vowel changes.

# 44 Velar palatalization before front vowels

## 44.1 Historical discussion

Luick places the change inside a broad early palatalizing movement. Under the heading "Frühe Verschiebungen in palatale Richtung," he treats English k and g before bright vowels together with the larger field of palatal effects (Luick 1914--1940, 157, §168). His emphasis falls on the environment first: velars before bright vowels and in the vicinity of the palatal glide belong to one early phonological sequence. The examples associated with that sequence, such as *ceaster* 'town', *geaf* 'gave', *giefan* 'give', and *giest* 'guest', already show that consonantal palatalization and later vowel effects stand close together historically, even when they must be distinguished analytically (Luick 1914--1940, 157–67, §§168–182).

Campbell narrows the picture by distinguishing plain velars from the especially palatal-prone sk cluster. His remark that "[sk] is more prone to palatalization and assibilation than [k]" is brief, but it makes clear that different members of the larger palatal field need not behave identically (Campbell 1959, 278, §440). Elsewhere in the same part of the grammar he uses forms such as *cild* 'child', *dæg* 'day', *giefan* 'give', and *giest* 'guest', which show how palatalized velars, palatal influence, and later umlautal outcomes meet in the same region of the lexicon without collapsing into one process (Campbell 1959, 69–72, 89, §§170, 190–191).

Hogg makes the conditioning sharper still. He states that the change takes place when the velar consonant is adjacent to and in the same syllable as a front vowel or the palatal consonant j (Hogg 1992, 103–4). This formulation replaces a broad list of palatal outcomes with a phonological environment defined by adjacency and syllable structure.

Ringe and Taylor make the chronological relation still clearer. When they write that "after initial velars and **sk* had been palatalized" West-Saxon diphthongization follows, plain velar palatalization becomes an earlier consonantal stage presupposed by later vowel developments (Ringe and Taylor 2014, 215, §6.5.1). Their own examples of the plain-velar rule, such as *weccan* 'wake', *licgan* 'lie', *lecgan* 'lay', *secg* 'retainer', *ecg* 'edge', *wicg* 'horse', and *brycg* 'bridge', illustrate the same point in lexical detail: front vowels and j create the palatal environment in which plain k and g cease to behave as plain velars (Ringe and Taylor 2014, 213–14, §6.4.1).

Luick describes a broad early movement; Campbell distinguishes plain velars from the sk complex; Hogg specifies the adjacency and syllable conditions; and Ringe and Taylor order the plain-velar change before West-Saxon diphthongization. Plain-velar palatalization thus forms part of a wider palatalizing environment without being identical to its neighboring changes.

## 44.2 SC052. Palatalization of **k* before front vowels and **j* (OEVelarPalatalizationKFront)

```
define OEVelarPalatalizationKFront [
    {*k} -> {*ʧ} || .#. _ EnglishStarFrontVowel,
    {*k} -> {*ʧ} || _ [{*i} | {*ī}],
    {*k} -> {*ʧ} || _ {*ī́},
    {*k} -> {*ʧ} || [{*i} | {*ī}] _ EnglishStarFrontVowel,
    {*k} -> {*ʧ} || {*ī́} _ EnglishStarFrontVowel,
    {*k} -> {*ʧ} || [{*i} | {*ī}] _ .#.,
    {*k} -> {*ʧ} || {*ī́} _ .#.
] .o. [
    {*k} {*k} -> {*ʧ} {*ʧ} || _ {*j}
] .o. [
    {*k} -> {*ʧ} || _ {*j}
] ;
```

The *weccan* 'wake', *licgan* 'lie', and *lecgan* 'lay' set identifies front vowels and j as the environment for palatalization of k (Ringe and Taylor 2014, 213–14, §6.4.1). These forms establish the conditioning; different witnesses establish the chronology.

Applied before Sievers-law syncope, PGmc **strákkijaną* ‘stretch’ yields †*streccan* rather than expected OE *streċċan* ‘stretch’. Applied after i-umlaut fronting, PGmc **kūi* ‘cow’ and **lúnganjō* ‘lungs’ yield *ċȳ* ‘cows’ and *lunġen* ‘lungs’ rather than expected OE *cȳ* ‘cows’ and *lungen* ‘lungs’. The front-vowel k change therefore follows Sievers-law syncope and precedes i-umlaut fronting.

## 44.3 SC052. Velar palatalization before front vowels (OEVelarPalatalization)

```
define OEVelarPalatalization [
    OEVelarPalatalizationKFront
] .o. [
    {*g} -> {*ʤ} || _ EnglishStarFrontVowel,
    {*g} -> {*ʤ} || EnglishStarFrontVowel _ .#.,
    {*g} -> {*ʤ} || EnglishStarFrontVowel _ EnglishStarFrontVowel,
    {*g} -> {*ʤ} || EnglishStarFrontVowel _ [EnglishStarConsonant - {*j}],
    {*g} {*g} -> {*ʤ} {*ʤ} || _ {*j}
] .o. [
    {*g} -> {*ʤ} || _ {*j}
];
```

Plain k and g palatalization in front-vocalic and j-adjacent environments follows sk-palatalization and occupies a sharply defined pre-umlaut interval. Applied before Sievers-law syncope, PGmc **strákkijaną* ‘stretch’ yields †*strecċan* rather than expected OE *streċċan* ‘stretch’. Applied after general i-umlaut, PGmc **kūi* ‘cow’ yields †*ċȳ* rather than expected *cȳ* ‘cows’, and PGmc **lúnganjō* ‘lungs’ yields †*lunġen* rather than expected *lungen* ‘lungs’. These witnesses place velar palatalization after Sievers-law syncope and before umlaut.

Luick, Campbell, and Ringe and Taylor place *cild* ‘child’ and *dæg* ‘day’ in a consonantal palatalization that precedes later vowel fronting (Luick 1914--1940, 157, §168; Campbell 1959, 278, §440; Ringe and Taylor 2014, 203–15, §§6.4.1, 6.5.1). The umlautal developments therefore receive plain k and g already reshaped beside front vowels and j.

The sk change belongs to the same palatalizing region with a separate scope. The *streċċan* ‘stretch’ evidence establishes a specific dependency on earlier syncope; it does not merge the two changes into one process.

# 45 Post-velar *w*-loss and loss of *w* before final *i*

## 45.1 Historical discussion

The first rule is a narrow loss of **w* after velars in the **ngw* sequence. Ringe and Taylor derive PGmc **singwan* 'sing' to Old English *singan* 'sing' (Ringe and Taylor 2014, 214, §6.4.2). This comparative evidence establishes the change, although no lexical evidence fixes its order relative to a neighboring rule.

The second rule is historically more legible. Campbell notes the recurring loss of **w* before **i* in unstressed position (Campbell 1959, 167, §406). Ringe and Taylor trace the development of *sǣ* 'sea' from earlier **saiwi-* / **sawi-* (Ringe and Taylor 2014, 257, §6.7.1), and Luick gives the same trajectory in his own historical grammar (Luick 1914--1940, 173, §187). The first rule is restricted to the **ngw* sequence; the second has a specific lexical witness and defined earlier and later limits.

## 45.2 SC053. Loss of *w* after velars (`OEPostVelarWLoss`)

```
define OEPostVelarWLoss [
    {*w} -> 0 || {*n} {*g} _
];
```

The comparative development `*singwan > singan` establishes narrow post-velar **w*-loss in the **ngw* sequence, yielding *singan* 'sing'. Moving SC053 OEPostVelarWLoss earlier or later leaves every output unchanged. Its pre-umlaut position therefore rests on comparative evidence, while the present lexicon supplies no neighboring boundary.

## 45.3 SC054. Loss of *w* before final *i* (`OEWLossBeforeI`)

```
define OEWLossBeforeI [
    {*w} -> 0 || EnglishStarVocalic _ {*i} .#.
];
```

The history of *sǣ* 'sea' explains why non-initial **w* disappeared before final unstressed **i*. Campbell describes the loss, Ringe and Taylor derive the form from **saiwi-*/**sawi-*, and Luick gives the parallel trajectory (Campbell 1959, 167, §406; Ringe and Taylor 2014, 257, §6.7.1; Luick 1914--1940, 173, §187). Loss of the glide allowed the preceding vowel to undergo the later fronting and lengthening.

The same witness supplies two distant limits. Before SC020 EAFFinalZDeletion or after SC063 OEHighVowelApocope, SC054 OEWLossBeforeI yields †*sǣw* rather than expected OE *sǣ* 'sea'. The loss must therefore follow final *z*-deletion and precede high-vowel apocope, while its exact position within that broad interval remains source-based.

# 46 The Old English i-umlaut and West Saxon palatal diphthongization

## 46.1 Historical discussion

Luick gives the change its traditional scale:

> Der wichtigste Fall von palataler Beeinflussung … war die Veränderung der urenglischen Vokale durch i oder j der Folgesilbe.
>
> (Luick 1914--1940, 166–67, §182)

Campbell gives the most compact classical formulation in English when he writes that "the process known as i-umlaut or i-mutation operates on practically all the sounds which it could theoretically affect in OE" (Campbell 1959, 69, §190). He immediately defines the core conditioning environment as a following `i` or `j`, and he goes on to trace the consequences across much of the vowel system, including forms such as *giest* 'guest', *giefan* 'give', *hierde* 'shepherd', and *ieldra* 'older' (Campbell 1959, 69–72, §§190–197).

Hogg continues in the same vein: "we come now to a change which is almost as uncontroversial as it is important" (Hogg 1992, 112). His examples, such as *bryd* 'bride', *trymman* 'strengthen', *bedd* 'bed', *ciest* 'chest', and *wiersa* 'worse', likewise emphasize that the change is a broad redistribution of vowel quality across the Old English vowel system (Hogg 1992, 112–14).

The narrower palatal-diphthongal material is described differently. Ringe and Taylor treat West-Saxon diphthongization after initial palatals as a distinct process (Ringe and Taylor 2014, 215, §6.5.1), and Fulk is even more explicit about its chronological delicacy when he calls it "diphthongization by initial palatal consonants (which precedes front umlaut but not breaking)" (Fulk 2018, 74, §4.13). Ringe and Taylor's examples such as *gieldan* 'pay', *scield* 'shield', and *scieppan* 'create' show that this narrower process is triggered by already palatal consonants and leads to specifically West-Saxon diphthongal outputs (Ringe and Taylor 2014, 215–16, §6.5.1).

Luick, Campbell, and Hogg treat i-umlaut as a system-wide change. Ringe and Taylor and Fulk distinguish from it a narrower West-Saxon process affecting words after initial palatals. The two changes act in different environments and produce different lexical consequences.

## 46.2 SC055. Fronting under i-umlaut (`OEIUmlautFronting`)

```
define OEIUmlautFronting [
    {*a} -> {*æ} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*ā} -> {*ǣ} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*e} -> {*i} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*o} -> {*e} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*ō} -> {*ē} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*u} -> {*y} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*ū} -> {*ȳ} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*á} -> {*æ} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*é} -> {*i} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*ó} -> {*e} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*ú} -> {*y} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger
];
```

The breadth of i-umlaut appears in lexical classes that share only a following high front vocoid. The forms *fylgan* 'follow', *gylden* 'golden', *wyrm* 'worm', and *giest* 'guest' exemplify the same `i`- or `j`-conditioned fronting across different vowels (Ringe and Taylor 2014, 222, §6.6.1; Campbell 1959, 69–72, §§190–191).

The cow and lung forms establish the lower boundary. If fronting precedes velar palatalization, PGmc **kūi* 'cow' yields †*ċȳ* rather than expected OE *cȳ* 'cows', and **lúnganjō* 'lungs' yields †*lunġen* rather than expected OE *lungen* 'lungs'. The consonantal change must therefore precede fronting.

The gift and sheath forms establish the upper boundary. If West Saxon palatal diphthongization precedes fronting, PGmc **géftiz* 'gift' yields †*ġieft* rather than expected OE *ġift* 'gift', and **skáiθiz* 'sheath' yields †*sċǣþ* rather than expected *sċēaþ* 'sheath'. Fronting consequently follows velar palatalization and precedes the West Saxon change; the other components of i-umlaut share those bounds.

## 46.3 SC055. Raising under i-umlaut (`OEIUmlautRaising`)

```
define OEIUmlautRaising [
    {*æ} -> {*e} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger
];
```

Raising of umlauted æ to e continues the same assimilatory event as fronting and therefore shares the chronology of general i-umlaut.

The same four forms fix both boundaries. If raising precedes velar palatalization, **kūi* 'cow' yields †*ċȳ* instead of expected *cȳ* 'cows' and **lúnganjō* 'lungs' yields †*lunġen* instead of expected *lungen* 'lungs'. If West Saxon palatal diphthongization precedes raising, **géftiz* 'gift' yields †*ġieft* rather than expected *ġift* 'gift', and **skáiθiz* 'sheath' yields †*sċǣþ* rather than expected *sċēaþ* 'sheath'. These forms place raising after velar palatalization and before West Saxon palatal diphthongization.

The sources do not describe umlaut as simple fronting alone. Campbell notes that the low front vowel changes again before `m` and `n` in most dialects (Campbell 1959, 69, §190), and Hogg likewise treats short front vowels as part of the same assimilatory system (Hogg 1992, 112).

## 46.4 SC055. Diphthongal outcomes under i-umlaut (`OEIUmlautDiphthong`)

```
define OEIUmlautDiphthong [
    {*ea} -> {*ie} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*ēa} -> {*īe} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*io} -> {*ie} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*īo} -> {*īe} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*eo} -> {*ie} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*ēo} -> {*īe} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*éa} -> {*íe} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*éo} -> {*íe} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger,
    {*ío} -> {*íe} || _ EnglishIUmlautIntervening EnglishIUmlautTrigger
];
```

Diphthongal outcomes belong to the same system-wide assimilation as simple-vowel fronting and raising. All three therefore belong to a single historical event.

The relevant examples are the recurring West-Saxon `ie` forms cited in the handbooks, including *giest* 'guest', *giefan* 'give', and *hierde* 'shepherd' in Campbell and *ciest* 'chest' in Hogg (Campbell 1959, 69–72, 78–80, §§190–191, 248–251; Hogg 1992, 112–14). These diphthongal outcomes form a distinct part of the general umlautal development alongside simple fronting.

The chronology comes from the cow/lung and gift/sheath contrasts. Placed before velar palatalization, diphthongal mutation over-palatalizes **kūi* 'cow' and **lúnganjō* 'lungs'; placed after West Saxon palatal diphthongization, it yields †*ġieft* and †*sċǣþ* instead of expected *ġift* 'gift' and *sċēaþ* 'sheath'. These failures place diphthongal mutation after velar palatalization and before West Saxon palatal diphthongization.

## 46.5 SC055. The composite i-umlaut rule (`OEIUmlaut`)

```
define OEIUmlaut OEIUmlautFronting
    .o. OEIUmlautRaising
```

```
    .o. OEIUmlautDiphthong;
```

The literature presents fronting, raising, and diphthongal mutation as effects of one historical development. They consequently occupy a single place in the Old English chronology.

The lower boundary is consonantal. If general umlaut precedes velar palatalization, PGmc **kūi* 'cow' yields †*ċȳ* rather than expected *cȳ* 'cows', and PGmc **lúnganjō* 'lungs' yields †*lunġen* rather than expected *lungen* 'lungs'. These over-palatalized forms place general umlaut after velar palatalization.

The upper boundary separates general umlaut from the narrower West Saxon process. If West Saxon palatal diphthongization precedes umlaut, PGmc **géftiz* 'gift' yields †*ġieft* rather than expected OE *ġift* 'gift', and **skáiθiz* 'sheath' yields †*sċǣþ* rather than expected *sċēaþ* 'sheath'. Together the two witness pairs place general umlaut after velar palatalization and before the West Saxon process.

## 46.6 SC056. West Saxon palatal diphthongization (OEWsPalatalDiphthongization)

```
define OEWsPalatalDiphthongization [
    {*æ} -> {*ea} || .#. [{*ʧ} | {*ʤ} | {*ʃ} | {*j}] _ [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant | .#.],
    {*ǣ} -> {*ēa} || .#. [{*ʧ} | {*ʤ} | {*ʃ} | {*j}] _ [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant | .#.],
    {*e} -> {*ie} || .#. [{*ʧ} | {*ʤ} | {*ʃ} | {*j}] _ [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant | .#.],
    {*ē} -> {*īe} || .#. [{*ʧ} | {*ʤ} | {*ʃ} | {*j}] _ [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant | .#.],
    {*é} -> {*íe} || .#. [{*ʧ} | {*ʤ} | {*ʃ} | {*j}] _ [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant | .#.],
    {*ḗ} -> {*īe} || .#. [{*ʧ} | {*ʤ} | {*ʃ} | {*j}] _ [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant | .#.]
];
```

West Saxon *gieldan* 'pay', *scield* 'shield', and *scieppan* 'create' show diphthongization after an already palatal consonant (Ringe and Taylor 2014, 215–16, §6.5.1). Their dialectal and phonological restriction separates this development from system-wide i-umlaut.

Hogg's *giefan* 'give' and *sceap* 'sheep' belong to the same palatal-consonant environment (Hogg 1992, 108–9). Fulk likewise assigns this diphthongization a place before front mutation and distinguishes the two processes (Fulk 2018, 74, §4.13).

The forms *ġift* 'gift' and *sċēaþ* 'sheath' fix the lower boundary. If West Saxon palatal diphthongization precedes general i-umlaut, PGmc **géftiz* 'gift' yields †*ġieft* rather than expected *ġift*, and PGmc **skáiθiz* 'sheath' yields †*sċǣþ* rather than expected *sċēaþ*. These witnesses place West Saxon palatal diphthongization after general umlaut; no tested lexical item supplies a later terminus ante quem.

The one-sided chronology reflects the difference in scale. General umlaut reorganizes the vowel system, whereas West Saxon palatal diphthongization affects a narrower dialectal class after palatal consonants. Its exact later placement remains undemonstrated by the present lexicon.

# 47 J-cluster coalescence

## 47.1 Historical discussion

Only a small lexical group reveals the coalescence of velars with **j*. Plain-velar and **sk* palatalization must already have run before **gj* and **kj* acquire their later outcomes. Campbell, Ringe and Taylor, and Fulk discuss the palatalized and fronted outcomes in *bīeġan* 'bend' and *sēċan* 'seek' without assigning this later cluster adjustment the status of a major sound law (Campbell 1959, 89, 107–8, §§170, 248–251; Ringe and Taylor 2014, 213–51, §§6.4.1, 6.5.1, 6.6.1–6.6.4; Fulk 2018, 65, 75, §§4.7, 4.13).

## 47.2 SC057. Coalescence of velar + **j* clusters (`OEJClusterCoalescence`)

```
define OEJClusterCoalescence (
    [{*g} {*j} -> {*ʤ}]
    .o. [{*k} {*j} -> {*ʧ}]
);
```

The forms *bīeġan* 'bend' and *sēċan* 'seek' determine the earlier boundary. If coalescence precedes SC052 OEVelarPalatalization, the developments behind *bīeġan* 'bend' and *sēċan* 'seek' are lost. Related forms such as *fylġan* 'follow', *heċġ* 'hedge', and *sengan* 'singe' fail in the same broader palatalization zone. PGmc **báugijaną* 'bow' yields †*bēaġan* rather than expected OE *bīeġan*, and PGmc **sōkijaną* 'seek' yields †*sōċan* rather than expected *sēċan*. This demonstrates that velar palatalization preceded coalescence. Nothing in the present lexicon supplies a terminus ante quem.

# 48 Back mutation

## 48.1 Historical discussion

West Saxon *giefan* ‘give’ and *wefan* ‘weave’ stand against non-West-Saxon *geofad* ‘gave’ and *weofan* ‘weave’. Ringe and Taylor use this contrast to define the dialectal profile of back mutation (Ringe and Taylor 2014, 319, §6.9.4). Campbell’s treatment of diphthongization before following back vowels includes *heofon* ‘heaven’ (Campbell 1959, 86, §207), while Hogg draws the instructive comparison with breaking (Hogg 1992, 112). Fulk accordingly separates back mutation from the earlier umlautal changes (Fulk 2018, 69, §4.8).

## 48.2 SC059. Back mutation before labials and liquids (`OEBackMutation`)

```
define OEBackMutation [
    {*e} -> {*eo} || _ [EnglishStarLabial | EnglishStarLiquid] {*u},
    {*æ} -> {*ea} || _ [EnglishStarLabial | EnglishStarLiquid]
     ↪  EnglishBackMutationTrigger,
    {*é} -> {*éo} || _ [EnglishStarLabial | EnglishStarLiquid] {*u}
];
```

Three witness forms bracket the chronology. If back mutation precedes SC048 OESecondaryNasalization, forms such as **gébaną* ‘give’ produce *ġeofan* ‘give’; the expected form is *ġiefan* ‘give’. **stélaną* ‘steal’ likewise produces *steolan* ‘steal’; the expected form is *stelan* ‘steal’. At the other edge, delaying back mutation until after SC078 OEWeakTailReduction makes **wébaną* ‘weave’ yield *weofan* ‘weave’; the expected form is *wefan* ‘weave’. Thus back mutation follows secondary nasalization but precedes the weak-tail reductions.

# 49 West Saxon palatal umlaut

## 49.1 Historical discussion

The reflexes *miht* ‘might’ and *niht* ‘night’ place West Saxon palatal umlaut after the principal umlautal developments. Campbell and Ringe and Taylor describe the forms themselves; Fulk supplies the broader chronology of palatal-vowel change (Campbell 1959, 107–8, §§248–251; Ringe and Taylor 2014, 215–51, §§6.5.1, 6.6.1–6.6.4; Fulk 2018, 65, 75, §§4.7, 4.13).

## 49.2 SC060. West Saxon palatal umlaut before **h*-clusters (OEWsPalatalUmlaut)

```
define OEWsPalatalUmlaut [
    {*eo} -> {*i} || _ OEHCluster .#.,
    {*io} -> {*i} || _ OEHCluster .#.,
    {*ie} -> {*i} || _ OEHCluster .#.,
    {*eo} -> {*i} || _ OEHCluster EnglishStarFrontVowel,
    {*io} -> {*i} || _ OEHCluster EnglishStarFrontVowel,
    {*ie} -> {*i} || _ OEHCluster EnglishStarFrontVowel,
    {*éo} -> {*i} || _ OEHCluster .#.,
    {*ío} -> {*i} || _ OEHCluster .#.,
    {*íe} -> {*i} || _ OEHCluster .#.,
    {*éo} -> {*i} || _ OEHCluster EnglishStarFrontVowel,
    {*ío} -> {*i} || _ OEHCluster EnglishStarFrontVowel,
    {*íe} -> {*i} || _ OEHCluster EnglishStarFrontVowel
];
```

The change to **i* before **h*-clusters can be ordered only on its earlier side. If palatal umlaut precedes SC055 OEIUmlaut, the forms behind *miht* ‘might’ and *niht* ‘night’ remain at the overdeveloped stage †*mieht* and †*nieht* rather than expected OE *miht* and *niht*. Consequently, i-umlaut precedes palatal umlaut. Reordering the latter against any tested later change leaves both witness forms unchanged.

# 50 Weak-tail nasal loss

## 50.1 Historical discussion

The pathway from **dōną* ‘do’ to *dōn* ‘do’ supplies the sole lexical thread through this reduction. Campbell, Hogg, and Fulk place such weak-tail losses among apocope and related late reductions (Campbell 1959, 144–45, §§345–349; Hogg 1992, 120–21; Fulk 2018, 91, §5.6). The witness, however, ties the change to a much older development. Its immediate neighbors remain untested.

## 50.2 SC061. Reduction of final nasal weak-tail endings (OEWeakTailNasalLoss)

```
define OEWeakTailNasalLoss [
    {*n} {*ą} -> {*n} || _ .#.,
    {*m} {*ą} -> {*m} || _ .#.
];
```

Final weak-tail **-ną* and **-mą* accordingly yield plain **-n* and **-m*.

Only *dōn* ‘do’ constrains the relative order. Placing this loss before the older n-stem loss makes the derivation record no output instead of expected OE *dōn* ‘do’. The older loss must therefore precede weak-tail nasal loss. Nothing in the current lexicon distinguishes among its possible later positions, and one witness cannot establish a wider historical development.

# 51 High-vowel apocope

## 51.1 Historical discussion

Final high vowels must survive long enough to condition umlaut before apocope removes them after heavy syllables and in the relevant trisyllabic patterns. Campbell, Hogg, Ringe and Taylor, and Fulk agree on this Old English development, though they differ over the extent of the surrounding syncope (Campbell 1959, 144–45, §§345–349; Hogg 1992, 120; Ringe and Taylor 2014, 284–303, §§6.8.1, 6.8.4; Fulk 2018, 91, §5.6).

## 51.2 SC063. High-vowel apocope after heavy syllables and in trisyllables (OEHighVowelApocope)

```
define OEHighVowelApocope [
    {*i} -> 0 || EnglishStarLongVowel OEAnyConsonant+ _ .#.,
    {*u} -> 0 || EnglishStarLongVowel OEAnyConsonant+ _ .#.,
    {*ų} -> 0 || EnglishStarLongVowel OEAnyConsonant+ _ .#.,
    {*i} -> 0 || EnglishStarLongDiphthong OEAnyConsonant+ _ .#.,
    {*u} -> 0 || EnglishStarLongDiphthong OEAnyConsonant+ _ .#.,
    {*ų} -> 0 || EnglishStarLongDiphthong OEAnyConsonant+ _ .#.,
    {*i} -> 0 || EnglishStarShortDiphthong OEAnyConsonant OEAnyConsonant+ _ .#.,
    {*u} -> 0 || EnglishStarShortDiphthong OEAnyConsonant OEAnyConsonant+ _ .#.,
    {*ų} -> 0 || EnglishStarShortDiphthong OEAnyConsonant OEAnyConsonant+ _ .#.,
    {*i} -> 0 || EnglishStarShortVowel OEAnyConsonant OEAnyConsonant+ _ .#.,
    {*u} -> 0 || EnglishStarShortVowel OEAnyConsonant OEAnyConsonant+ _ .#.,
    {*ų} -> 0 || EnglishStarShortVowel OEAnyConsonant OEAnyConsonant+ _ .#.,
    {*i} -> 0 || EnglishStarLongVowel OEAnyConsonant+ EnglishStarShortVowel
     ↪  OEAnyConsonant+ _ .#.,
    {*u} -> 0 || EnglishStarLongVowel OEAnyConsonant+ EnglishStarShortVowel
     ↪  OEAnyConsonant+ _ .#.,
    {*ų} -> 0 || EnglishStarLongVowel OEAnyConsonant+ EnglishStarShortVowel
     ↪  OEAnyConsonant+ _ .#.,
    {*i} -> 0 || EnglishStarLongDiphthong OEAnyConsonant+ EnglishStarShortVowel
     ↪  OEAnyConsonant+ _ .#.,
    {*u} -> 0 || EnglishStarLongDiphthong OEAnyConsonant+ EnglishStarShortVowel
     ↪  OEAnyConsonant+ _ .#.,
    {*ų} -> 0 || EnglishStarLongDiphthong OEAnyConsonant+ EnglishStarShortVowel
     ↪  OEAnyConsonant+ _ .#.,
    {*i} -> 0 || EnglishStarShortDiphthong OEAnyConsonant OEAnyConsonant+
     ↪  EnglishStarShortVowel OEAnyConsonant+ _ .#.,
    {*u} -> 0 || EnglishStarShortDiphthong OEAnyConsonant OEAnyConsonant+
     ↪  EnglishStarShortVowel OEAnyConsonant+ _ .#.,
    {*ų} -> 0 || EnglishStarShortDiphthong OEAnyConsonant OEAnyConsonant+
     ↪  EnglishStarShortVowel OEAnyConsonant+ _ .#.,
    {*i} -> 0 || EnglishStarShortDiphthong OEAnyConsonant EnglishStarShortVowel
     ↪  OEAnyConsonant+ _ .#.,
    {*u} -> 0 || EnglishStarShortDiphthong OEAnyConsonant EnglishStarShortVowel
     ↪  OEAnyConsonant+ _ .#.,
    {*ų} -> 0 || EnglishStarShortDiphthong OEAnyConsonant EnglishStarShortVowel
     ↪  OEAnyConsonant+ _ .#.,
    {*i} -> 0 || EnglishStarShortVowel OEAnyConsonant EnglishStarShortVowel
     ↪  OEAnyConsonant+ _ .#.,
    {*u} -> 0 || EnglishStarShortVowel OEAnyConsonant EnglishStarShortVowel
     ↪  OEAnyConsonant+ _ .#.,
    {*ų} -> 0 || EnglishStarShortVowel OEAnyConsonant EnglishStarShortVowel
     ↪  OEAnyConsonant+ _ .#.,
    {*i} -> 0 || EnglishStarShortVowel OEAnyConsonant OEAnyConsonant+
     ↪  EnglishStarShortVowel OEAnyConsonant+ _ .#.,
    {*u} -> 0 || EnglishStarShortVowel OEAnyConsonant OEAnyConsonant+
     ↪  EnglishStarShortVowel OEAnyConsonant+ _ .#.,
    {*ų} -> 0 || EnglishStarShortVowel OEAnyConsonant OEAnyConsonant+
     ↪  EnglishStarShortVowel OEAnyConsonant+ _ .#.,
```

```
    {*i} -> 0 || EnglishStarLongVowel _ .#.,
    {*u} -> 0 || {*x} _ .#.,
    {*ų} -> 0 || {*x} _ .#.,
    {*i} -> 0 || {*x} _ .#.
];
```

Final **i*, **u*, and **ų* cannot disappear before completing their umlautal work. Applied before SC055 OEIUmlaut, PGmc **kūi* ‘cow’ yields †*cū* rather than expected OE *cȳ* ‘cow’, and PGmc **brūdiz* ‘bride’ yields †*brūd* rather than expected OE *brȳd* ‘bride’. Conversely, if apocope waits until after SC072 OEUnstressedLongVowelShortening, PGmc **fúrxtīnaz* ‘fright’ yields †*fyrht* rather than expected OE *fyrhte* ‘fright’. The three witnesses establish the sequence i-umlaut, high-vowel apocope, unstressed long-vowel shortening.

# 52 Post-apocope **n*-loss and medial syncope

## 52.1 Historical discussion

Evidence for post-apocope reduction is strikingly uneven. The inherited feminine *in*-stem represented by Gothic *faurhtei*, OE *fyrhtu*, and oblique OE *fyrhte* supplies the relevant evidence (Orel 2003, 120; Ringe and Taylor 2014, 380–81; Campbell 1959, 236, §589.7). No comparable witness orders the medial syncope that follows. Hogg, Ringe and Taylor, and Fulk describe both processes within the late history of weak syllables (Hogg 1992, 120–21; Ringe and Taylor 2014, 264–303, §§6.7.3–6.8.4; Fulk 2018, 91, §5.6).

## 52.2 SC064. Loss of stem-final **n* after long **ī* (`NWGmcInStemNLoss`)

```
define NWGmcInStemNLoss [{*n} -> 0 || {*ī} _ .#.];
```

Only final **n* after long **ī* is at issue, as in the inherited *in*-stem behind OE *fyrhte* ‘fright’.

CAPR models the oblique OE form through the Proto-Germanic genitive singular **fúrxtīnaz* ‘fright’, following the project convention of using an appropriate non-nominative paradigm cell when the nominative does not supply the required derivation. Within this selected genitive derivation, the same input fixes both ordering boundaries. Before SC041 PWGmcFinalBareALoss, PGmc **fúrxtīnaz* ‘fright’ yields †*fyrhten* rather than expected OE *fyrhte* ‘fright’. After SC072 OEUnstressedLongVowelShortening, PGmc **fúrxtīnaz* again yields †*fyrhten* rather than expected *fyrhte* ‘fright’. I therefore order final bare-a loss, stem-final n-loss, and unstressed long-vowel shortening in that sequence. Both boundaries are firm within the selected genitive derivation and depend on one inherited lexeme/paradigm.

## 52.3 SC065. Medial syncope before dentals after heavy syllables (OEMedialSyncope)

Loss of medial **i* before dentals belongs to the late weak-tail history described by Hogg, Ringe and Taylor, and Fulk (Hogg 1992, 120–21; Ringe and Taylor 2014, 264–303, §§6.7.3–6.8.4; Fulk 2018, 91, §5.6).

```
define OEMedialSyncope [
    {*i} -> 0 || EnglishStarLongVowel OEAnyConsonant+ _ [{*θ}|{*ð}|{*d}|{*t}],
    {*i} -> 0 || EnglishStarLongDiphthong OEAnyConsonant+ _ [{*θ}|{*ð}|{*d}|{*t}],
    {*i} -> 0 || EnglishStarShortVowel OEAnyConsonant OEAnyConsonant+ _
     ↪  [{*θ}|{*ð}|{*d}|{*t}]
];
```

No diagnostic word establishes a local chronology. Moving medial syncope to either end of the tested range leaves every output unchanged. Its handbook placement after apocope and before later cluster simplification therefore remains preferable, but the present lexicon cannot demonstrate it.

# 53 Late syncope and degemination

## 53.1 Historical discussion

Vowel loss creates the clusters upon which later assimilation and degemination operate. Hogg and Ringe and Taylor describe this dependence, while Brunner's *netle* 'nettle' beside later *netele* 'nettle' supplies a concrete lexical type (Hogg 1992, 120–21; Ringe and Taylor 2014, 264–96, §§6.7.3–6.8.2; Brunner 1965, 144–45, §§158–159). Fulk places this syncope after i-umlaut (Fulk 2018, 91, §5.6).

The three relations are not equally secure. Lexical evidence orders syncope and degemination; the intervening dental assimilation has no independent ordering witness.

## 53.2 SC066. L-adjacent syncope in medial syllables (OELAdjacentSyncope)

```
define OELAdjacentSyncope [
    {*i} -> 0 || EnglishStarShortVowel OEAnyConsonant+ _ {*l},
    {*i} -> 0 || EnglishStarLongVowel OEAnyConsonant+ _ {*l},
    {*i} -> 0 || EnglishStarDiphthong OEAnyConsonant+ _ {*l}
];
```

The loss of medial **i* before **l* is late enough to preserve earlier umlaut, as *netle* 'nettle' and *spinl* 'spindle' demonstrate.

Placed before i-umlaut, PGmc **nátilōn* 'nettle' yields †*nætle* rather than expected OE *netle* 'nettle', and PGmc **spénnilō* 'spindle' yields †*spenl* rather than expected *spinl* 'spindle'. Placed after preconsonantal degemination, PGmc **spénnilō* yields †*spinnl* rather than expected *spinl*. The witnesses therefore establish the sequence i-umlaut, l-adjacent syncope, preconsonantal degemination. The first relation separates two historical phases; the second is a direct feeding relation, since syncope creates the cluster that degemination simplifies.

## 53.3 SC067. Dental assimilation in newly formed clusters (OEDentalAssimilation)

```
define OEDentalAssimilation [
    {*θ} -> 0 || {*t} _
];
```

Loss of **θ* after **t* resolves a dental cluster produced by syncope (Hogg 1992, 120–21; Ringe and Taylor 2014, 279–96, §§6.7.5, 6.8.2). No witness distinguishes its position: moving dental assimilation across every tested neighbor leaves the outputs unchanged. I nevertheless place it after syncope, which supplies its input, and before the more general cluster simplification described in the handbooks. This order is phonologically motivated, not established by a lexical contrast.

## 53.4 SC068. Preconsonantal degemination before sonorants (OEPreconsonantalDegemination)

```
define OEPreconsonantalDegemination OEPreconsonantalDegemTT .o.
 ↪  OEPreconsonantalDegemNN;
```

Preconsonantal **tt* and **nn* simplify only after syncope has created a following sonorant cluster, as in *spinl* 'spindle' (Ringe and Taylor 2014, 279–96, §§6.7.5, 6.8.2).

Placed before l-adjacent syncope, PGmc **spénnilō* 'spindle' yields †*spinnl* rather than expected OE *spinl* 'spindle'. Syncope must therefore create the cluster before degemination simplifies it.

Reordering degemination against any tested later change leaves the witness unchanged, so no terminus ante quem is known.

# 54 Early o-shortening

## 54.1 Historical discussion

After the principal palatal and umlautal changes, unstressed vowels undergo shortening, fronting, merger, and sometimes complete loss. Campbell describes the early shortening of unaccented long vowels, while Hogg, Ringe and Taylor, and Fulk relate it to apocope, syncope, and the later reductions (Campbell 1959, 148, §355; Hogg 1992, 120–21; Ringe and Taylor 2014, 298–314, §§6.8.3–6.9.3; Fulk 2018, 90–96, §§5.6–5.7).

Early o-shortening has only a distant earlier boundary. The rules that follow, especially SC070 OEUnstressedFrontingEarly and SC072 OEUnstressedLongVowelShortening, have more closely defined relations.

## 54.2 SC069. Early shortening of unstressed **ō* before nasals (OEEarlyOShortening)

```
define OEEarlyOShortening [
    {*ō} -> {*a} || EnglishStarVocalic [EnglishStarConsonant |
    ↪  EnglishPalatalConsonant]+ _ EnglishStarNasal
];
```

The rule shortens unstressed long **ō* before a following nasal. Because this shortening happens early, the resulting **a* can still participate in the later fronting and merger that shape many weak final syllables.

Moving the rule before SC023 PNWGmcNStemNLoss, PGmc **nḗdrōn* ‘adder’ yields †*nǣdran* rather than expected OE *nǣdre* ‘adder’, PGmc **érθōn* ‘earth’ yields †*eorþan* rather than expected *eorþe* ‘earth’, and PGmc **fláskōn* ‘flask’ yields †*flascan* rather than expected *flasce* ‘flask’. The same earlier shift also disrupts forms such as *heorte* ‘heart’ and *līne* ‘line’. This broad set of failures requires SC069 OEEarlyOShortening to follow SC023 PNWGmcNStemNLoss.

If the rule is moved later within the tested sequence, no output differs from the expected one. The lexical evidence therefore does not identify a corresponding later constraint. The sources place early **ō*-shortening before the later weak-tail changes without fixing a closer local order.

# 55 Early unstressed fronting and later o-shortening

## 55.1 Historical discussion

Campbell distinguishes the shortening of unaccented long vowels, while Hogg, Ringe and Taylor, and Fulk place fronting and shortening within a later history of syncope and final-vowel adjustment (Campbell 1959, 148, §355; Hogg 1992, 120–21; Ringe and Taylor 2014, 298–314, §§6.8.3–6.9.3; Fulk 2018, 90–96, §§5.6–5.7). Earlier unstressed fronting precedes later o-shortening.

SC070 OEUnstressedFrontingEarly has both an earlier and a later lexical breakpoint. SC071 OELateOShortening confirms their reciprocal order, but no lexical evidence fixes its later boundary.

## 55.2 SC070. Early fronting of unstressed **a* (OEUnstressedFrontingEarly)

```
define OEUnstressedFrontingEarly OEUnstressedAFronting;
```

The rule fronts unstressed **a* to **æ* after the earlier shortening has created a frontable vowel but before the later shortening of unstressed **ō*. It produces endings such as OE *-en* in *lungen* 'lungs'.

If the rule is moved before SC052 OEVelarPalatalization, PGmc **lúnganjō* 'lungs' yields †*lunġen* rather than expected OE *lungen* 'lungs'. If the rule is delayed until after SC071 OELateOShortening, PGmc **búrōθi* 'bears' yields †*boreþ* rather than expected OE *boraþ* 'bears', and PGmc **mḗnōθz* 'month' yields †*mōneþ* rather than expected *mōnaþ* 'month'. The witness forms require SC070 OEUnstressedFrontingEarly to follow SC052 OEVelarPalatalization and precede SC071 OELateOShortening.

The relation to SC071 OELateOShortening is local. The earlier boundary at SC052 OEVelarPalatalization places fronting after the older palatal developments.

## 55.3 SC071. Later shortening of unstressed **ō* (`OELateOShortening`)

The following rule handles the later shortening stage.

```
define OELateOShortening [
    {*ō} -> {*a} || EnglishStarVocalic [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant]+ _ [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant]*
];
```

The rule shortens the remaining unstressed long **ō* after fronting, producing the later "stable a" endings in OE *boraþ* 'bears' and *liornaþ* 'learns'.

Moving the rule before SC070 OEUnstressedFrontingEarly makes PGmc **búrōθi* 'bears' yield †*boreþ* rather than expected OE *boraþ* 'bears', and PGmc **líznōθi* 'learns' yield †*liorneþ* rather than expected *liornaþ* 'learns'. The contrast requires SC071 OELateOShortening to follow SC070 OEUnstressedFrontingEarly. Moving it later within the tested range creates no equally sharp failure.

# 56 Unstressed long-vowel shortening and ae-merger

## 56.1 Historical discussion

Campbell describes the shortening of unaccented long vowels, and Ringe and Taylor place it among the last prehistoric Old English changes before the merger of unstressed **æ* with **e* (Campbell 1959, 148, §355; Hogg 1992, 120–21; Ringe and Taylor 2014, 298–314, §§6.8.3–6.9.3; Fulk 2018, 90–96, §§5.6–5.7).

SC072 OEUnstressedLongVowelShortening and SC073 OEUnstressedAEMerger have a reciprocal ordering relation. SC064 NWGmcInStemNLoss supplies the earlier boundary of shortening, and SC085 OEHLoss the later boundary of the merger.

## 56.2 SC072. Shortening of unstressed long vowels (OEUnstressedLongVowelShortening)

```
define OEUnstressedLongVowelShortening OEUnstressedLongVowelShortening1
    .o. OEUnstressedLongVowelShortening2
    .o. OEUnstressedLongVowelShortening3
    .o. OEUnstressedLongVowelShortening5
    .o. OEUnstressedLongVowelShortening6
    .o. OEUnstressedLongVowelShortening7
    .o. OEUnstressedLongVowelShortening8;
```

The rule shortens the remaining unstressed long vowels before weak final syllables reach their later forms. A small group of lexical witnesses fixes its chronology.

If the rule is moved before SC064 NWGmcInStemNLoss, PGmc **fúrxtīnaz* ‘fright’ yields †*fyrhten* rather than expected OE *fyrhte* ‘fright’. If the rule is delayed until after SC073 OEUnstressedAEMerger, PGmc **nḗdrōn* ‘adder’ yields †*nǣdræ* rather than expected OE *nǣdre* ‘adder’, and PGmc **fádēr* ‘father’ yields †*fædær* rather than expected *fæder* ‘father’. These outputs require SC072 OEUnstressedLongVowelShortening to follow SC064 NWGmcInStemNLoss and precede SC073 OEUnstressedAEMerger.

Shortening therefore follows the earlier weak-tail preparation and immediately precedes the merger.

## 56.3 SC073. Merger of unstressed **æ* with **e* (`OEUnstressedAEMerger`)

The following rule handles the merger stage.

```
define OEUnstressedAEMerger OEWeakTailReduction3;
```

The rule merges unstressed **æ* with **e* after shortening has produced the weak final vowels, yielding the ordinary OE *-e* spellings.

Its earlier and later relations are both concrete. If the rule is moved before SC072 OEUnstressedLongVowelShortening, PGmc **nḗdrōn* ‘adder’ yields †*nǣdræ* rather than expected OE *nǣdre* ‘adder’, and PGmc **fádēr* ‘father’ yields †*fædær* rather than expected *fæder* ‘father’. If the rule is delayed until after SC085 OEHLoss, PGmc **táixōn* ‘toe’ yields †*tāæ* rather than expected OE *tā* ‘toe’. These failures show that SC072 OEUnstressedLongVowelShortening must come before SC073 OEUnstressedAEMerger, and that SC073 OEUnstressedAEMerger must come before SC085 OEHLoss.

The lexical evidence fixes the local order after SC072 OEUnstressedLongVowelShortening and places the merger before the later h-loss and contraction.

# 57 Medial unstressed-i lowering

## 57.1 Historical discussion

Hogg and Ringe and Taylor treat the late weakening and merger of unstressed vowels as a continuing history (Hogg 1992, 120–21; Ringe and Taylor 2014, 327–32, §§6.9.5–6.9.6). SC074 OEMedUnstressedILowering1 lowers medial unstressed *i*; SC075 OEMedUnstressedILowering preserves *i* before **ng* in words of the *sċilling* 'shilling' type.

General lowering precedes the restricted restoration before **ng*. The evidence is narrower than that for SC072 OEUnstressedLongVowelShortening and SC073 OEUnstressedAEMerger.

## 57.2 SC074. First medial unstressed-*i* lowering (OEMedUnstressedILowering1)

```
define OEMedUnstressedILowering1 [
    {*i} -> {*e} || EnglishStarVocalic [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant]+ _
];
```

The rule lowers medial unstressed **i* to **e* after a preceding vocalic syllable. The resulting *e*-outcome is reversed before **ng*.

If the rule is moved before SC072 OEUnstressedLongVowelShortening, PGmc **fúrxtīnaz* 'fright' yields †*fyrhti* rather than expected OE *fyrhte* 'fright'. If it is delayed until after SC075 OEMedUnstressedILowering, PGmc **skíllingaz* 'shilling' yields †*sċilleng* rather than expected *sċilling* 'shilling'. The derivations require SC074 OEMedUnstressedILowering1 to follow SC072 OEUnstressedLongVowelShortening and precede SC075 OEMedUnstressedILowering.

The evidence is narrow on each side. The rule follows unstressed long-vowel shortening and precedes the more specific **ng* preservation.

## 57.3 SC075. Preservation of medial unstressed **i* before **ng* (OEMedUnstressedILowering)

The following rule reverses the lowering before **ng*.

```
define OEMedUnstressedILowering [
    {*e} -> {*i} || _ {*n} {*g}
];
```

The rule restores **i* before **ng*, preventing the broader lowering from producing the wrong medial vowel in forms such as *sċilling* 'shilling'.

Moving the rule before SC074 OEMedUnstressedILowering1 makes PGmc **skíllingaz* 'shilling' yield †*sċilleng* rather than expected OE *sċilling* 'shilling'. On this evidence, I take SC075 OEMedUnstressedILowering to follow SC074 OEMedUnstressedILowering1. Moving it later within the tested range creates no equally sharp failure.

# 58 Prefix i-reduction

## 58.1 Historical discussion

Late weak-tail reduction affects unstressed prefixes as well as inflectional endings and medial vowels. Fulk's discussion of prefix vowels accounts for OE **be-* and **ne-* (Fulk 2018, 97, §5.7). Hogg and Ringe and Taylor place such weakening within the broader late history of unstressed vowels (Hogg 1992, 120–21; Ringe and Taylor 2014, 298–332, §§6.8.3–6.9.6).

The tested forms do not determine the rule's position relative to a neighboring change.

## 58.2 SC076. Reduction of prefixal **i* in unstressed position (OEPrefixIReduction)

```
define OEPrefixIReduction [
    {*i} -> {*ĕ} || .#. [{*b} | {*n}] _ [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant] EnglishStarVocalic
];
```

The rule reduces unstressed prefixal **i* to a weaker vowel in the *bi-* and *ni-* type prefixes before a consonant plus a following vowel. The development accounts for later prefix spellings such as OE **be-* and **ne-*.

If the rule is moved earlier or later within the tested sequence, no output differs from the expected one. The lexical evidence therefore does not place SC076 OEPrefixIReduction before or after any specific neighboring change.

The handbooks attest late prefix-vowel weakening, but the precise placement remains approximate. No lexical failure fixes it.

# 59 Weak-tail reduction

## 59.1 Historical discussion

Campbell, Hogg, Ringe and Taylor, and Fulk describe a late history in which apocope, shortening, contraction, and further weak-tail reductions reshape final syllables (Campbell 1959, 148, §355; Hogg 1992, 120–21; Ringe and Taylor 2014, 298–314, §§6.8.3–6.9.3; Fulk 2018, 90–91, §5.6). Lexical failures place the remaining weak-tail reduction after unstressed fronting and before contraction.

## 59.2 SC078. Reduction of remaining weak-tail vowels (OEWeakTailReduction)

```
define OEWeakTailReduction OEWeakTailReduction1;
```

The rule reduces the remaining weak-tail vowels, preventing a broad class of *-en* and extra-vowel outcomes.

I place the change after SC070 OEUnstressedFrontingEarly and before SC086 OEContraction. Moving it before SC070 OEUnstressedFrontingEarly, PGmc **bákaną* 'bake' yields †*bacen* rather than expected OE *bacan* 'bake', and PGmc **bíndaną* 'bind' yields †*binden* rather than expected *bindan* 'bind', alongside a much wider set of comparable *-en* failures. If the rule is delayed until after SC086 OEContraction, PGmc **fléuxaną* 'flee' yields †*flēoan* rather than expected OE *flēon* 'flee', and PGmc **sláxaną* 'slay' yields †*sleaan* rather than expected *slēan* 'slay'.

The earlier boundary spans a wide interval and does not establish a close neighboring relation. The later boundary is narrower: SC078 OEWeakTailReduction precedes SC086 OEContraction.

# 60 Final-j loss and final geminate simplification

## 60.1 Historical discussion

After SC079 OEJLossAfterHeavy removes **j* in heavy environments, forms such as *lungen* ‘lungs’ acquire a final geminate. SC080 OEFinalGeminateSimplification then removes the second nasal.

SC079 OEJLossAfterHeavy has a broad earlier boundary at SC055 OEIUmlaut. SC080 OEFinalGeminateSimplification is fixed only by the final *nn* outcome in the following derivation.

## 60.2 SC079. Loss of **j* after heavy syllables (`OEJLossAfterHeavy`)

```
define OEJLossAfterHeavy [
    {*j} -> 0 || (EnglishStarLongVowel | EnglishStarDiphthong)
     ↪  [EnglishStarConsonantNoR | EnglishPalatalConsonant] _,
    {*j} -> 0 || EnglishStarShortVowel [EnglishStarConsonant |
     ↪  EnglishPalatalConsonant] [EnglishStarConsonantNoR |
     ↪  EnglishPalatalConsonant] _
];
```

The rule removes **j* after the relevant heavy-syllable configurations, after the earlier umlaut-sensitive vocalism has developed. The affected glide is **j*.

If the rule is moved before SC055 OEIUmlaut, PGmc **galáubijaną* ‘believe’ yields †*ġelēafan* rather than expected OE *ġelīefan* ‘believe’, PGmc **báugijaną* ‘bow’ yields †*bēaġan* rather than expected *bīeġan* ‘bow’, and PGmc **fúlgijaną* ‘follow’ yields †*fulġan* rather than expected *fylġan* ‘follow’. If it is delayed until after SC080 OEFinalGeminateSimplification, PGmc **lúnganjō* ‘lungs’ yields †*lungenn* rather than expected OE *lungen* ‘lungs’. I accordingly take SC079 OEJLossAfterHeavy to follow SC055 OEIUmlaut and precede SC080 OEFinalGeminateSimplification.

The earlier boundary is broad, but the relation to final geminate simplification is local.

## 60.3 SC080. Simplification of final geminates (OEFinalGeminateSimplification)

The following rule handles the final simplification directly.

```
define OEFinalGeminateSimplification [
    {*n} -> 0 || {*n} _ .#.
];
```

The rule removes the extra final nasal in forms where the preceding derivation has already created a final geminate.

Moving the rule before SC079 OEJLossAfterHeavy makes PGmc **lúnganjō* ‘lungs’ yield †*lungenn* rather than expected OE *lungen* ‘lungs’. These failures require SC080 OEFinalGeminateSimplification to follow SC079 OEJLossAfterHeavy. Moving it later within the tested range before SC087 OERMetathesis creates no new failure.

# 61 J-strengthening, vocalization, and ei-contraction

## 61.1 Historical discussion

SC081 OEJStrengtheningAfterFrontDiphthong preserves a consonantal outcome after front diphthongs. SC082 OEIntervocalicJVocalization then vocalizes the remaining intervocalic **j*, and SC083 OEUnstressedEIContraction removes the resulting *ei*-like sequence in weak verbal endings.

The output of each rule conditions the next. SC082 OEIntervocalicJVocalization has local lexical evidence on both sides; SC081 OEJStrengtheningAfterFrontDiphthong has a distant earlier boundary, and SC083 OEUnstressedEIContraction has no tested later boundary.

## 61.2 SC081. Strengthening of **j* after front diphthongs (OEJStrengtheningAfterFrontDiphthong)

```
define OEJStrengtheningAfterFrontDiphthong [
    {*j} -> {*ʒ} || [{*ēa}|{*ḗa}|{*íe}|{*īe}|{*éa}] _ EnglishStarVocalic
];
```

After the relevant front diphthongs, **j* first strengthened to a consonantal outcome; otherwise it would have vocalized too early.

If the rule is moved before SC055 OEIUmlaut, PGmc **stráwjaną* 'strew' yields †*strēaġan* rather than expected OE *strīeġan* 'strew'. If it is delayed until after SC082 OEIntervocalicJVocalization, the same PGmc form yields †*strīeian* rather than *strīeġan*. The order test requires SC081 OEJStrengtheningAfterFrontDiphthong to follow SC055 OEIUmlaut and precede SC082 OEIntervocalicJVocalization.

The earlier constraint reaches back to SC055 OEIUmlaut and therefore defines a wide interval. The *strīeġan* 'strew' derivation fixes the local relation to SC082 OEIntervocalicJVocalization.

## 61.3 SC082. Intervocalic vocalization of **j* (OEIntervocalicJVocalization)

```
define OEIntervocalicJVocalization [
    {*j} -> {*i} || EnglishStarVocalic _ EnglishStarVocalic
];
```

The rule vocalizes intervocalic **j* to **i*, creating the *ei*-like sequence later removed by SC083 OEUnstressedEIContraction in many weak verb forms.

Moving the rule before SC081 OEJStrengtheningAfterFrontDiphthong makes PGmc **stráwjaną* 'strew' yield †*strīeian* rather than expected OE *strīeġan* 'strew'. Delaying it until after SC083 OEUnstressedEIContraction makes PGmc **búrōjaną* 'bore' yield †*boreian* rather than expected OE *borian* 'bore', PGmc **xándlōjaną* 'handle' yield †*handleian* rather than expected *handlian* 'handle', and PGmc **mákōjaną* 'make' yield †*maceian* rather than expected *macian* 'make'. The witness forms require SC082 OEIntervocalicJVocalization to follow SC081 OEJStrengtheningAfterFrontDiphthong and precede SC083 OEUnstressedEIContraction.

SC082 OEIntervocalicJVocalization is therefore ordered between strengthening and contraction.

## 61.4 SC083. Contraction of unstressed *ei* (`OEUnstressedEIContraction`)

The final rule removes the extra unstressed *e* before *i*.

```
define OEUnstressedEIContraction [
    {*e} -> 0 || EnglishStarVocalic [EnglishStarConsonant |
    ↪  EnglishPalatalConsonant]+ _ {*i}
```

```
];
```

The rule contracts the unstressed *ei*-like sequence that the preceding vocalization would otherwise leave behind in forms such as *borian* 'bore' and *liccian* 'lick'.

Moving the rule before SC082 OEIntervocalicJVocalization makes PGmc **búrōjanq̨* 'bore' yield †*boreian* rather than expected OE *borian* 'bore', PGmc **líznōjaną* 'learn' yield †*liorneian* rather than expected *liornian* 'learn', and PGmc **líkkōjaną* 'lick' yield †*licceian* rather than expected *liccian* 'lick'. The contrast requires SC083 OEUnstressedEIContraction to follow SC082 OEIntervocalicJVocalization. Moving it later within the tested range before SC087 OERMetathesis creates no new failure.

# 62 H-loss and contraction

## 62.1 Historical discussion

When SC085 OEHLoss removes intervocalic **h*, it creates hiatus. SC086 OEContraction immediately resolves the resulting vowel sequence.

Ringe and Taylor describe this late sequence of *h*-loss and contraction (Ringe and Taylor 2014, 305–14, §§6.9.1–6.9.3). Fulk places the contracted verbs in a broader Germanic context (Fulk 2018, 270, §12.21), and Luick describes the corresponding West Germanic contractions (Luick 1914--1940, 165).

## 62.2 SC085. Loss of intervocalic **h* (`OEHLoss`)

```
define OEHLoss [
    {*x} -> 0 || EnglishStarVocalic _ EnglishStarVocalic
];
```

The rule removes intervocalic **h*, creating the hiatus that later contraction must resolve.

If the rule is moved before SC073 OEUnstressedAEMerger, PGmc **táixōn* 'toe' yields †*tāæ* rather than expected OE *tā* 'toe'. If it is delayed until after SC086 OEContraction, PGmc **fléuxaną* 'flee' yields †*flēoan* rather than expected OE *flēon* 'flee', PGmc **sláxaną* 'slay' yields †*sleaan* rather than expected *slēan* 'slay', PGmc **téxun* 'draw' yields †*teoon* rather than expected *tēon* 'draw', and PGmc **táixōn* yields †*tāe* rather than expected *tā*. These outputs require SC085 OEHLoss to follow SC073 OEUnstressedAEMerger and precede SC086 OEContraction.

The earlier boundary rests on one witness; the four later witnesses establish the immediate relation to contraction.

## 62.3 SC086. Contraction of the resulting hiatus (`OEContraction`)

The following rule contracts the hiatus left by SC085 OEHLoss.

```
define OEContraction [
    {*a} {*a} -> {*ā},
    {*e} {*e} -> {*ē},
    {*i} {*i} -> {*ī},
    {*o} {*o} -> {*ō},
    {*u} {*u} -> {*ū},
    {*ea} {*a} -> {*ēa},
    {*ēa} {*a} -> {*ēa},
    {*eo} {*a} -> {*ēo},
    {*ēo} {*a} -> {*ēo},
    {*eo} {*o} -> {*ēo},
    {*ēo} {*o} -> {*ēo},
    {*éo} {*o} -> {*ḗo},
    {*ḗo} {*o} -> {*ḗo},
    {*ā} {*a} -> {*ā},
    {*ā} {*e} -> {*ā},
    {*ē} {*a} -> {*ē},
    {*ē} {*e} -> {*ē},
    {*ḗ} {*a} -> {*ḗ},
    {*ḗ} {*e} -> {*ḗ},
    {*ī} {*a} -> {*ī},
    {*ī} {*e} -> {*ī},
    {*ī́} {*a} -> {*ī́},
    {*ī́} {*e} -> {*ī́},
    {*ō} {*a} -> {*ō},
    {*ō} {*e} -> {*ō},
    {*ū} {*a} -> {*ū},
```

```
    {*ū} {*e} -> {*ū}
];
```

The rule contracts the vowel sequences created after *h*-loss, producing *flēon* ‘flee’, *slēan* ‘slay’, and *tēon* ‘draw’.

Moving contraction before SC085 OEHLoss makes PGmc **fléuxaną* ‘flee’ yield †*flēoan* rather than expected OE *flēon* ‘flee’, PGmc **sláxaną* ‘slay’ yield †*sleaan* rather than expected *slēan* ‘slay’, PGmc **téxun* ‘draw’ yield †*teoon* rather than expected *tēon* ‘draw’, and PGmc **táixōn* ‘toe’ yield †*tāe* rather than expected *tā* ‘toe’. The derivations require SC086 OEContraction to follow SC085 OEHLoss. Moving it later within the tested range before SC087 OERMetathesis creates no new failure. The more distant SC078 OEWeakTailReduction relation establishes only that weak-tail reduction precedes contraction.

# 63 R-metathesis

## 63.1 Historical discussion

Sievers-Brunner describes r-metathesis in forms such as *berstan* ‘burst’, *forst* ‘frost’, and *cærse* ‘cress’ (Brunner 1965, 159, §179). Luick likewise treats it as a later rearrangement whose interaction with breaking remains variable (Luick 1914--1940, 201).

The evidence establishes that breaking precedes metathesis. It does not establish an ordering relation between SC086 OEContraction and SC087 OERMetathesis.

## 63.2 SC087. Metathesis of **r* with a following short vowel (OERMetathesis)

```
define OERMetathesis [
    {*r} {*e} -> {*e} {*r} || EnglishStarConsonant _ {*s} {*t},
    {*r} {*u} -> {*u} {*r} || EnglishStarConsonant _ {*s} {*t},
    {*r} {*i} -> {*i} {*r} || EnglishStarConsonant _ {*s} {*t},
    {*r} {*o} -> {*o} {*r} || EnglishStarConsonant _ {*s} {*t},
    {*r} {*a} -> {*a} {*r} || EnglishStarConsonant _ {*s} {*t},
    {*r} {*é} -> {*é} {*r} || EnglishStarConsonant _ {*s} {*t},
    {*r} {*ó} -> {*ó} {*r} || EnglishStarConsonant _ {*s} {*t},
    {*r} {*á} -> {*á} {*r} || EnglishStarConsonant _ {*s} {*t}
];
```

The rule moves **r* across a following short vowel in the relevant late clusters, producing forms such as *berstan* ‘burst’ where an earlier order would still show a broken vowel sequence.

Moving the rule before SC044 OEBreaking makes PGmc **bréstaną* ‘burst’ yield †*beorstan* rather than expected OE *berstan* ‘burst’. On this evidence, I take SC087 OERMetathesis to follow SC044 OEBreaking. Moving it later within the tested sequence alters no output.

The lexical evidence fixes the earlier relation but does not identify a corresponding later constraint. The sources treat r-metathesis as a late rearrangement after breaking without placing it immediately beside contraction.

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